Topics in Functional Analysis

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Preface

This is the lecture note for the graduate course *Functional Analysis* taught by Prof. Yutian Li in 2023 Fall at the Chinese University of Hong Kong, Shenzhen. The note is edited and revised with the support of ChatGPT. The materials presume a basic knowledge of functional analysis, so contents that are close to the boundary of these prerequisites may be omitted. Topics of the note include: review of linear functional analysis, weak convergence, spectral theory, semigroups, Sobolev spaces, and calculus of variations.

Contents

Chapter 1 Review of the Great Theorems

1.1 Hahn–Banach theory

Theorem 1.1.1 (Hahn–Banach theorem over real vector spaces)**.** Let X be a real vector space, let $Y \subseteq X$ be a linear subspace, and let $p\colon X \to \mathbb{R}$ be sublinear:

$$p(\alpha x) = \alpha p(x) \quad (\alpha \geq 0), \qquad p(x+y) \leq p(x) + p(y).$$

If $f\colon Y \to \mathbb{R}$ is linear and $f(y) \leq p(y)$ for every $y \in Y$, then there is a linear functional $\widetilde{f}\colon X \to \mathbb{R}$ such that

$$\widetilde{f}|_Y = f, \qquad \widetilde{f}(x) \leq p(x) \quad (x \in X).$$

Proof. Partially order the set of dominated linear extensions (Z, g) of (Y, f) by extension. Every chain has the upper bound obtained by taking the union of its domains and functionals, so Zorn's lemma gives a maximal extension (M, g). If $M \neq X$, choose $x_0 \notin M$. An extension to $M + \mathbb{R}x_0$ has the form

$$g_\alpha(m + tx_0) = g(m) + t\alpha.$$

The domination condition is equivalent to

$$\sup_{m \in M} \big(g(m) - p(m - x_0)\big) \leq \alpha \leq \inf_{m \in M} \big(p(m + x_0) - g(m)\big).$$

Sublinearity of p shows that the left endpoint is no larger than the right one. Choosing such an α contradicts maximality. Hence $M = X$. □

Corollary 1.1.2 (Norm-preserving Hahn–Banach extension)**.** Let X be a normed space, let $Y \subseteq X$ be a linear subspace, and let $f \in Y^*$. There is $\widetilde{f} \in X^*$ satisfying

$$\widetilde{f}|_Y = f, \qquad \|\widetilde{f}\| = \|f\|.$$

Proof. Apply Hahn–Banach with $p(x) = \|f\|\,\|x\|$. The domination gives $\|\widetilde{f}\| \leq \|f\|$, while restriction to Y gives the reverse inequality. □

Remark 1.1.3 (A useful uniqueness criterion)**.** If X^* is strictly convex, then every norm-preserving Hahn–Banach extension is unique. Indeed, if F_1 and F_2 are two distinct extensions of f with $\|F_1\| = \|F_2\| = \|f\| > 0$, then $(F_1 + F_2)/2$ is another extension and must have norm at least $\|f\|$; this contradicts strict convexity after normalization. The converse is not automatic from the one-line argument in the source and is therefore not asserted here without an additional global unique-extension hypothesis.

Proposition 1.1.4 (Norming functionals)**.** Let X be a normed space and let $x \in X \setminus \{0\}$. There is $f_x \in X^*$ such that

$$f_x(x) = \|x\|, \qquad \|f_x\| = 1.$$

Consequently, continuous linear functionals separate points: if $x \neq y$, then some $f \in X^*$ satisfies $f(x) \neq f(y)$.

Proof. Define $g(tx) = t\|x\|$ on $\operatorname{span}\{x\}$. Then $\|g\| = 1$ and a norm-preserving Hahn–Banach extension gives f_x. Apply this to $x - y$ for the separation statement. □

1.2 Geometric separation

Lemma 1.2.1 (Minkowski functional)**.** Let X be a normed space and let $C \subseteq X$ be open and convex with $0 \in C$. Define

$$p_C(x) = \inf\{t > 0 : x \in tC\}.$$

Then p_C is finite and sublinear. Moreover, for some $M > 0$,

$$p_C(x) \leq M\|x\| \quad (x \in X), \qquad C = \{x \in X : p_C(x) < 1\}.$$

Proof. Choose $r > 0$ with $B(0, r) \subseteq C$. Then $x \in tC$ for every $t > \|x\|/r$, so $p_C(x) \leq \|x\|/r$. Positive homogeneity follows directly from the definition. If $a > p_C(x)$ and $b > p_C(y)$, then $x/a, y/b \in C$ and convexity yields

$$\frac{x+y}{a+b} = \frac{a}{a+b}\frac{x}{a} + \frac{b}{a+b}\frac{y}{b} \in C.$$

Letting $a \downarrow p_C(x)$ and $b \downarrow p_C(y)$ proves subadditivity. Finally, openness and convexity give $C = \{p_C < 1\}$. □

Lemma 1.2.2 (Separation of an open convex set and a point)**.** Let $C \subseteq X$ be open and convex and let $x_0 \notin C$. Then there are $f \in X^* \setminus \{0\}$ and $\alpha \in \mathbb{R}$ such that

$$f(x) < \alpha = f(x_0) \qquad (x \in C).$$

Proof. Translate so that $0 \in C$. Let p_C be the Minkowski functional and define $g(tx_0) = t$ on $\operatorname{span}\{x_0\}$. Since $p_C(x_0) \geq 1$, one checks that $g \leq p_C$ on this line; for negative t this follows from $g(tx_0) \leq 0 \leq p_C(tx_0)$. Hahn–Banach gives $f \leq p_C$ on X. Thus $f(x_0) = 1$ and $f(x) < 1$ for $x \in C$. □

Theorem 1.2.3 (Geometric Hahn–Banach separation)**.** Let A, B be nonempty disjoint convex subsets of a normed space X.

(a). If A is open, then there are $f \in X^* \setminus \{0\}$ and $\alpha \in \mathbb{R}$ such that

$$f(a) < \alpha \leq f(b) \qquad (a \in A,\ b \in B).$$

(b). If A is closed and B is compact, then there are $f \in X^* \setminus \{0\}$, $\alpha \in \mathbb{R}$, and $\varepsilon > 0$ such that

$$f(a) + \varepsilon \le \alpha \le f(b) \qquad (a \in A,\ b \in B).$$

Proof. For the first assertion, $A - B$ is open and convex and does not contain 0. Separating 0 from $A - B$ gives $f(a-b) < 0$, hence $\sup_A f \le \inf_B f$, with strict inequality on the open side.
For the second assertion, $A - B$ is closed because B is compact, and $0 \notin A - B$. Hence $B(0,r) \cap (A - B) = \varnothing$ for some $r > 0$. Apply the first assertion to the open ball and the convex set $A - B$. The resulting functional is bounded away from zero on $A - B$, which gives the stated strict gap after rescaling and changing sign if necessary. □

1.3 Baire category and uniform boundedness

Definition 1.3.1. A subset N of a topological space is *nowhere dense* if $\operatorname{int}(\overline{N}) = \varnothing$. A space is of the *first category* if it is a countable union of nowhere dense sets, and of the *second category* otherwise.

Lemma 1.3.2. For a subset A of a topological space X, the following are equivalent:

(a). A is dense in X;

(b). every nonempty open subset of X meets A;

(c). $\operatorname{int}(X \setminus A) = \varnothing$.

Theorem 1.3.3 (Baire category theorem)**.** Every nonempty complete metric space is of the second category. Equivalently:

(a). a countable union of closed sets with empty interior has empty interior;

(b). a countable intersection of open dense sets is dense.

Proof. Let (U_n) be open dense subsets of a complete metric space X, and let $O \subseteq X$ be nonempty and open. Choose a closed ball $\overline{B}(x_1, r_1) \subseteq O \cap U_1$. Inductively choose

$$\overline{B}(x_{n+1}, r_{n+1}) \subseteq B(x_n, r_n) \cap U_{n+1}, \qquad 0 < r_{n+1} \le \tfrac{1}{2} r_n.$$

Then (x_n) is Cauchy. If $x_n \to x$, the closed-ball nesting gives $x \in \overline{B}(x_n, r_n) \subseteq U_n$ for every n, and also $x \in O$. Thus $O \cap \bigcap_n U_n \ne \varnothing$, so the intersection is dense. Taking complements gives the closed-set formulation. □

Example 1.3.4. A normed space with a countably infinite Hamel basis cannot be complete: it is the union of the finite-dimensional spans of the first n basis vectors, each of which is closed and has empty interior. Baire's theorem also underlies the existence of continuous nowhere differentiable functions on $[0,1]$.

Theorem 1.3.5 (Uniform boundedness principle)**.** Let X be a Banach space, let Y be a normed space, and let $\{T_i : i \in I\} \subseteq \mathcal{B}(X, Y)$. If

$$\sup_{i \in I} \|T_i x\| < \infty \qquad (x \in X),$$

then

$$\sup_{i\in I} \|T_i\| < \infty.$$

Proof. Set

$$F_n = \{x \in X : \sup_{i\in I} \|T_i x\| \le n\}.$$

Each F_n is closed and $X = \bigcup_n F_n$. Baire's theorem gives $x_0 \in X$ and $r > 0$ with $B(x_0, r) \subseteq F_N$ for some N. If $\|h\| < r$, then $x_0 \pm h \in F_N$, and therefore

$$2\|T_i h\| = \|T_i(x_0 + h) - T_i(x_0 - h)\| \le 2N.$$

Taking $h = (r/2)x$ with $\|x\| \le 1$ gives $\|T_i x\| \le 2N/r$ uniformly in i. □

Theorem 1.3.6 (Banach–Steinhaus pointwise-limit theorem)**.** Let X be Banach, let Y be normed, and let $T_n \in \mathcal{B}(X, Y)$. Suppose $T_n x$ converges for every $x \in X$, and define $Tx = \lim_n T_n x$. Then $T \in \mathcal{B}(X, Y)$ and

$$\|T\| \le \liminf_{n\to\infty} \|T_n\|.$$

Proof. Linearity is inherited from the pointwise limit. For every x, the sequence $(T_n x)$ is bounded, so uniform boundedness yields $M := \sup_n \|T_n\| < \infty$. Hence $\|Tx\| \le M\|x\|$. More precisely, for $\|x\| = 1$,

$$\|Tx\| = \lim_n \|T_n x\| \le \liminf_n \|T_n\|,$$

and taking the supremum over the unit sphere gives the claim. □

Theorem 1.3.7 (Weak boundedness implies norm boundedness)**.** Let $B \subseteq X$, where X is a normed space. If $f(B)$ is bounded in $\mathbb{K}$ for every $f \in X^*$, then B is norm bounded.

Proof. For $b \in B$, define $J_b \in (X^*)^*$ by $J_b(f) = f(b)$. The family $\{J_b : b \in B\}$ is pointwise bounded on the Banach space X^*, so uniform boundedness gives $\sup_{b\in B} \|J_b\| < \infty$. Hahn–Banach gives $\|J_b\| = \|b\|$, and the result follows. □

1.4 Open mapping, bounded inverse, and equivalent norms

Definition 1.4.1. A map between topological spaces is *open* if it sends open sets to open sets.

Theorem 1.4.2 (Banach open mapping theorem)**.** Let X and Y be Banach spaces and let $T \in \mathcal{B}(X, Y)$ be surjective. Then T is an open map.

Proof. It is enough to show that $T(B_X)$ contains a neighbourhood of 0. Since

$$Y = \bigcup_{n=1}^{\infty} T(nB_X),$$

Baire's theorem gives n such that $\overline{T(nB_X)}$ has nonempty interior. Taking differences shows that, for some $\delta > 0$,

$$B_Y(0, \delta) \subseteq \overline{T(B_X)}.$$

After rescaling, assume $B_Y(0,1) \subseteq \overline{T(B_X)}$. Given $y \in B_Y(0,1/2)$, choose inductively $x_k \in 2^{-k} B_X$ so that

$$\left\| y - T \sum_{j=1}^{k} x_j \right\| < 2^{-k-1}.$$

The series $\sum_k x_k$ converges to some x with $\|x\| < 1$, and continuity of T gives $Tx = y$. Thus $B_Y(0,1/2) \subseteq T(B_X)$. Translation and scaling now show that every open set has open image. $\square$

Corollary 1.4.3 (Bounded inverse theorem)**.** A bounded bijective linear map between Banach spaces has a bounded inverse.

Proof. The open mapping theorem says that T^{-1} is continuous at 0, hence bounded. $\square$

Example 1.4.4 (A priori estimate for a boundary-value problem)**.** Let

$$X = \{u \in C^2[0,1] : u(0) = u(1) = 0\}, \quad \|u\|_X = \|u\|_\infty + \|u'\|_\infty + \|u''\|_\infty,$$

and let $Y = C[0,1]$ with the supremum norm. For continuous coefficients a, b, c, the operator

$$Lu = au'' + bu' + cu$$

is bounded from X to Y. If the boundary-value problem $Lu = f$ has a unique solution $u \in X$ for every $f \in Y$—equivalently, if L is bijective—then the bounded inverse theorem gives a constant C such that

$$\|u\|_X \leq C\|Lu\|_\infty.$$

The bijectivity hypothesis is essential; it does not follow merely from the formula for L.

Theorem 1.4.5 (Equivalence of complete norms)**.** Let $\|\cdot\|_1$ and $\|\cdot\|_2$ be norms making the same vector space X Banach. If

$$\|x\|_2 \leq C\|x\|_1 \qquad (x \in X),$$

then there is $C' > 0$ such that

$$\|x\|_1 \leq C'\|x\|_2 \qquad (x \in X).$$

Thus the two norms are equivalent and induce the same topology.

Proof. The identity map $(X, \|\cdot\|_1) \to (X, \|\cdot\|_2)$ is a bounded bijection between Banach spaces. Apply the bounded inverse theorem. $\square$

1.5 Closed operators and the closed graph theorem

Definition 1.5.1. For a linear operator $T\colon \mathrm{D}(T) \subseteq X \to Y$, its graph is

$$\mathrm{Graph}(T) = \{(x, Tx) : x \in \mathrm{D}(T)\} \subseteq X \times Y.$$

The operator is *closed* if its graph is closed. Equivalently, whenever $x_n \in \mathrm{D}(T)$, $x_n \to x$ in X, and $Tx_n \to y$ in Y, one has $x \in \mathrm{D}(T)$ and $Tx = y$.

Remark 1.5.2. Every bounded linear operator is closed. The converse fails when the domain is not complete. For example,

$$D\colon C^1[0,1] \subset C[0,1] \to C[0,1], \qquad Du = u',$$

is closed when both spaces use the supremum norm, but it is unbounded.

Theorem 1.5.3 (Banach closed graph theorem)**.** Let X and Y be Banach spaces. An everywhere-defined linear map $T\colon X \to Y$ is bounded if and only if it is closed.

Proof. Only the closed-to-bounded direction needs proof. Equip X with the graph norm

$$\|x\|_T = \|x\|_X + \|Tx\|_Y.$$

Closedness of T implies that $(X, \|\cdot\|_T)$ is complete. The identity map

$$(X, \|\cdot\|_T) \longrightarrow (X, \|\cdot\|_X)$$

is a bounded bijection, so its inverse is bounded. Hence $\|x\|_T \leq C\|x\|_X$, and therefore $\|Tx\|_Y \leq (C-1)\|x\|_X$. □

Theorem 1.5.4 (Everywhere-defined symmetric maps)**.** Let H be a real Hilbert space and let $T\colon H \to H$ be a map satisfying

$$\langle Tx, y\rangle = \langle x, Ty\rangle \qquad (x, y \in H).$$

Then T is linear and bounded. The same conclusion holds over a complex Hilbert space when the inner product is taken linear in the first variable and the symmetric identity $\langle Tx, y\rangle = \langle x, Ty\rangle$ is imposed.

Proof. For $x_1, x_2, y \in H$ and scalars a, b,

$$\langle T(ax_1 + bx_2) - aTx_1 - bTx_2, y\rangle = \langle ax_1 + bx_2, Ty\rangle - a\langle x_1, Ty\rangle - b\langle x_2, Ty\rangle = 0.$$

Thus T is linear. If $x_n \to x$ and $Tx_n \to z$, then for every $y \in H$,

$$\langle z, y\rangle = \lim_n \langle Tx_n, y\rangle = \lim_n \langle x_n, Ty\rangle = \langle x, Ty\rangle = \langle Tx, y\rangle.$$

Hence $z = Tx$, so T is closed. The closed graph theorem gives boundedness. □

1.6 Adjoints, compact maps, and closed ranges

Definition 1.6.1. For normed spaces X, Y and $T \in \mathcal{B}(X, Y)$, the adjoint $T^*\colon Y^* \to X^*$ is defined by

$$(T^*y^*)(x) = y^*(Tx).$$

A bounded linear map $T\colon X \to Y$ is *compact* if it maps bounded sets to relatively compact sets, equivalently, if every bounded sequence (x_n) has a subsequence for which (Tx_{n_k}) converges in Y.

Theorem 1.6.2 (Schauder theorem)**.** A bounded linear operator $T\colon X \to Y$ is compact if and only if its adjoint $T^*\colon Y^* \to X^*$ is compact.

Remark 1.6.3. A proof is given in Chapter 6, where compact operators are studied systematically.

Theorem 1.6.4 (Closed range theorem)**.** Let X, Y be Banach spaces and $T \in \mathcal{B}(X, Y)$. Then the following are equivalent:

(a). $\operatorname{Ran}(T)$ is norm closed in Y;

(b). $\operatorname{Ran}(T^*)$ is norm closed in X^*;

(c). $\operatorname{Ran}(T^*)$ is weak-$*$ closed in X^*.

Whenever these conditions hold,

$$\operatorname{Ran}(T) = {}^{\perp}\operatorname{Ker}(T^*), \qquad \operatorname{Ran}(T^*) = \operatorname{Ker}(T)^{\perp}.$$

Without a closed-range hypothesis one always has

$$\overline{\operatorname{Ran}(T)} = {}^{\perp}\operatorname{Ker}(T^*), \qquad \overline{\operatorname{Ran}(T^*)}^{w^*} = \operatorname{Ker}(T)^{\perp}.$$

Corollary 1.6.5. For $T \in \mathcal{B}(X, Y)$ between Banach spaces, the following are equivalent:

(a). T is surjective;

(b). there is $c > 0$ such that $\|T^*y^*\| \geq c\|y^*\|$ for every $y^* \in Y^*$;

(c). T^* is injective and has closed range.

Proof. If T is surjective, its induced map $X/\operatorname{Ker} T \to Y$ is a bounded bijection, so the bounded inverse theorem yields the lower bound on T^*. Conversely, the lower bound makes T^* injective with closed range. The closed range theorem then gives $\overline{\operatorname{Ran}(T)} = {}^{\perp}\operatorname{Ker}(T^*) = Y$, while $\operatorname{Ran}(T)$ is closed, so $\operatorname{Ran}(T) = Y$. $\square$

Chapter 2 Weak Convergence and Weak Topology

2.1 Topologies generated by maps

Definition 2.1.1. A *topology* τ on a set X is a family of subsets of X such that $\varnothing, X \in \tau$, arbitrary unions of members of τ belong to τ, and finite intersections of members of τ belong to τ. The pair (X, τ) is a topological space.

A family $\mathcal{B} \subseteq \tau$ is a *base* if every open set is a union of members of $\mathcal{B}$. A family $\mathcal{S}$ is a *subbase* if the finite intersections of members of $\mathcal{S}$ form a base.

Definition 2.1.2. A map $F\colon (X, \tau_X) \to (Y, \tau_Y)$ is continuous if $F^{-1}(O) \in \tau_X$ for every $O \in \tau_Y$. A sequence (x_n) converges to x if, for every neighbourhood U of x, one has $x_n \in U$ for all sufficiently large n.

Definition 2.1.3 (Initial topology)**.** Let X be a set and let $\phi_i\colon X \to (Y_i, \tau_i)$ be maps indexed by $i \in I$. The *initial topology* generated by the family (ϕ_i) is the topology with subbase

$$\{\phi_i^{-1}(O) : i \in I,\ O \in \tau_i\}.$$

It is the weakest topology on X for which every ϕ_i is continuous.

Proposition 2.1.4. For the initial topology generated by (ϕ_i):

(a). $x_n \to x$ if and only if $\phi_i(x_n) \to \phi_i(x)$ for every i;

(b). a map $\psi\colon Z \to X$ is continuous if and only if every $\phi_i \circ \psi$ is continuous.

Proof. The forward implications follow from continuity of the coordinate maps. For the converse in the first statement, a basic neighbourhood of x is a finite intersection

$$\bigcap_{j=1}^{m} \phi_{i_j}^{-1}(O_j), \qquad \phi_{i_j}(x) \in O_j,$$

and coordinatewise convergence eventually places x_n in every factor. The second statement follows because inverse images commute with finite intersections and arbitrary unions. □

2.2 The weak topology

Definition 2.2.1. Let X be a normed space. The *weak topology* $\sigma(X, X^*)$ is the initial topology generated by all $f \in X^*$. A sequence converges weakly, written $x_n \rightharpoonup x$, if

$$f(x_n) \longrightarrow f(x) \qquad (f \in X^*).$$

Theorem 2.2.2. The weak topology on a normed space is Hausdorff.

Proof. If $x \neq y$, Hahn–Banach gives $f \in X^*$ with $f(x) \neq f(y)$. Disjoint open intervals about these two scalar values have disjoint inverse images containing x and y. □

Proposition 2.2.3 (Basic weak neighbourhoods)**.** For $x_0 \in X$, the sets

$$V(x_0; f_1, \dots, f_m; \varepsilon) = \{x \in X : |f_j(x - x_0)| < \varepsilon,\ 1 \leq j \leq m\}$$

form a neighbourhood base at x_0. Allowing different positive radii in the coordinates gives the same topology.

Theorem 2.2.4 (Finite-dimensional case)**.** If X is finite dimensional, weak convergence and norm convergence coincide. Equivalently, the weak and norm topologies on X are the same.

Proof. Choose a basis $e_1, \dots, e_d$ and its coordinate functionals $e_1^*, \dots, e_d^*$. If $x_n \rightharpoonup x$, then every coordinate of $x_n - x$ converges to zero. All norms on the finite-dimensional coordinate space are equivalent, so $\|x_n - x\| \to 0$. □

Example 2.2.5. If (e_n) is an orthonormal sequence in a Hilbert space, then $e_n \rightharpoonup 0$: Bessel's inequality implies $\langle x, e_n \rangle \to 0$ for every x. In contrast, in ℓ^1 weak convergence of sequences is equivalent to norm convergence (Schur's theorem).

Proposition 2.2.6 (Uniqueness, boundedness, and lower semicontinuity)**.** Let X be a normed space.

(a). Weak limits are unique.

(b). If $x_n \rightharpoonup x$, then (x_n) is norm bounded.

(c). If $x_n \rightharpoonup x$, then

$$\|x\| \leq \liminf_{n \to \infty} \|x_n\|.$$

Proof. Uniqueness follows because the weak topology is Hausdorff. For boundedness, consider $J_{x_n} \in (X^*)^*$, $J_{x_n}(f) = f(x_n)$. Weak convergence makes $(J_{x_n}(f))$ bounded for each f, so uniform boundedness and $\|J_{x_n}\| = \|x_n\|$ give $\sup_n \|x_n\| < \infty$.

For the last assertion, choose $f \in X^*$ with $\|f\| = 1$ and $f(x) = \|x\|$ (or, in the complex case, rotate a norming functional). Then

$$\|x\| = \lim_n f(x_n) \leq \liminf_n \|x_n\|.$$

□

Theorem 2.2.7 (Weak closure of the unit sphere). Let X be infinite dimensional and let $S_X = \{x : \|x\| = 1\}$. Then

$$\overline{S_X}^{w} = B_X = \{x : \|x\| \leq 1\}.$$

Consequently, the weak topology is strictly weaker than the norm topology.

Proof. The closed unit ball is weakly closed because the norm is weakly lower semicontinuous, so $\overline{S_X}^{w} \subseteq B_X$. Conversely, let $\|x_0\| < 1$ and let $V = V(x_0; f_1, \ldots, f_m; \varepsilon)$ be a basic weak neighbourhood. The finite-codimensional space $\bigcap_{j=1}^{m} \operatorname{Ker} f_j$ is nonzero; choose $0 \neq y$ in it. The continuous function $t \mapsto \|x_0 + ty\|$ starts below 1 and tends to infinity, so for some t it equals 1. Since all $f_j(y)$ vanish, $x_0 + ty \in V \cap S_X$. □

2.3 Weak continuity and compactness

Proposition 2.3.1. Let X, Y be normed spaces.

(a). If $A \in \mathcal{B}(X, Y)$ and $x_n \rightharpoonup x$, then $Ax_n \rightharpoonup Ax$.

(b). If $A \in \mathcal{K}(X, Y)$ and $x_n \rightharpoonup x$, then $\|Ax_n - Ax\| \to 0$.

(c). If $B \colon X \times Y \to Z$ is a bounded bilinear map, then $x_n \to x$ and $y_n \to y$ in norm imply $B(x_n, y_n) \to B(x, y)$ in norm.

Proof. For the first assertion, $g(Ax_n) = (A^*g)(x_n) \to (A^*g)(x)$ for every $g \in Y^*$. For the second, weak convergence makes (x_n) bounded. If $Ax_n \not\to Ax$, a subsequence stays a fixed distance away. Compactness gives a further norm-convergent subsequence, while the first assertion forces its weak limit to be Ax; hence its norm limit is Ax, a contradiction. The bilinear estimate

$$\|B(x_n, y_n) - B(x, y)\| \leq \|B\| \big(\|x_n - x\| \, \|y_n\| + \|x\| \, \|y_n - y\| \big)$$

proves the third assertion. □

Theorem 2.3.2 (Mazur's theorem). If $x_n \rightharpoonup x$ in a normed space, then for every n there is a finite convex combination of the tail,

$$y_n \in \operatorname{conv}\{x_k : k \geq n\},$$

such that $\|y_n - x\| \to 0$.

Proof. Let $C_n = \overline{\operatorname{conv}\{x_k : k \geq n\}}^{\|\cdot\|}$. If $x \notin C_n$, the geometric Hahn–Banach theorem gives $f \in X^*$ and α with $f(x) > \alpha \geq f(z)$ for $z \in C_n$. But $f(x_k) \to f(x)$, contradicting the inequality for all $k \geq n$. Thus $x \in C_n$, and one may choose $y_n \in \operatorname{conv}\{x_k : k \geq n\}$ with $\|y_n - x\| < 1/n$. □

Corollary 2.3.3 (Banach–Saks–Mazur block form). If $x_n \rightharpoonup x$, one can choose integers $1 \leq m_1 < m_2 < \cdots$ and convex combinations

$$z_j \in \operatorname{conv}\{x_k : m_j \leq k < m_{j+1}\}$$

such that $z_j \to x$ in norm. In particular, weak convergence always admits strongly convergent convex blocks.

Theorem 2.3.4 (Convex closures)**.** For a convex subset C of a normed space,

$$\overline{C}^{\|\cdot\|} = \overline{C}^{w}.$$

Hence a convex set is norm closed if and only if it is weakly closed.

Proof. The weak topology is coarser, so norm closure is contained in weak closure. If $x \notin \overline{C}^{\|\cdot\|}$, separate the closed convex set $\overline{C}^{\|\cdot\|}$ from x by a continuous linear functional. The separating weakly open half-space excludes x from the weak closure. □

Corollary 2.3.5. A convex norm-lower-semicontinuous function $\varphi \colon X \to (-\infty, \infty]$ is weakly lower semicontinuous.

Proof. Every sublevel set $\{x : \varphi(x) \leq a\}$ is convex and norm closed, hence weakly closed. □

Theorem 2.3.6 (Weak continuity of linear maps)**.** Let E and F be Banach spaces and let $T \colon E \to F$ be linear. Then T is norm continuous if and only if it is continuous from $\sigma(E, E^*)$ to $\sigma(F, F^*)$.

Proof. Norm continuity implies weak continuity because $f \circ T \in E^*$ for every $f \in F^*$. Conversely, weak continuity makes the graph weakly closed in $E \times F$. A weakly closed set is norm closed, so the closed graph theorem implies that T is bounded. □

2.4 The Weak-Star Topology

Definition 2.4.1. The *weak-$*$ topology* $\sigma(E^*, E)$ on E^* is the initial topology generated by the evaluation maps

$$\phi_x(f) = f(x), \qquad x \in E.$$

Thus $f_\alpha \xrightarrow{*} f$ if and only if $f_\alpha(x) \to f(x)$ for every $x \in E$.

Proposition 2.4.2. The weak-$*$ topology is Hausdorff. A neighbourhood base at $f_0 \in E^*$ is given by

$$V(f_0; x_1, \ldots, x_m; \varepsilon) = \{f \in E^* : |f(x_j) - f_0(x_j)| < \varepsilon,\ 1 \leq j \leq m\}.$$

Moreover,

$$\text{norm convergence} \implies \text{weak convergence} \implies \text{weak-}*\text{ convergence}$$

for nets in E^*, after identifying the weak topology as $\sigma(E^*, E^{**})$.

Lemma 2.4.3 (Finite functional dependence)**.** Let $\ell, \ell_1, \ldots, \ell_m$ be linear functionals on a vector space V. If

$$\bigcap_{j=1}^{m} \operatorname{Ker} \ell_j \subseteq \operatorname{Ker} \ell,$$

then ℓ belongs to $\operatorname{span}\{\ell_1, \ldots, \ell_m\}$.

Proof. Define $L\colon V \to \mathbb{K}^m$ by $L(v) = (\ell_1(v), \dots, \ell_m(v))$. The kernel inclusion makes $\ell(v)$ depend only on $L(v)$, so it induces a linear functional on the subspace $L(V) \subseteq \mathbb{K}^m$. Extend this finite-dimensional functional to $\mathbb{K}^m$ and read off its coordinates. $\square$

Theorem 2.4.4 (Dual of the weak-$*$ dual)**.** A linear functional $\Lambda\colon E^* \to \mathbb{K}$ is weak-$*$ continuous if and only if there is an $x \in E$ such that

$$\Lambda(f) = f(x) \qquad (f \in E^*).$$

In other words, the continuous dual of $(E^*, \sigma(E^*, E))$ is the canonical copy of E.

Proof. Evaluations are continuous by definition. Conversely, continuity at 0 gives $x_1, \dots, x_m \in E$ and $\varepsilon > 0$ such that $|f(x_j)| < \varepsilon$ for all j implies $|\Lambda(f)| < 1$. By homogeneity, $\bigcap_j \operatorname{Ker} \phi_{x_j} \subseteq \operatorname{Ker} \Lambda$. The finite functional lemma gives $\Lambda = \sum_j a_j \phi_{x_j} = \phi_{\sum_j a_j x_j}$. $\square$

Corollary 2.4.5. A hyperplane in E^* is weak-$*$ closed if and only if it has the form

$$\{f \in E^* : f(x) = a\}$$

for some $x \in E$ and $a \in \mathbb{K}$.

2.5 Banach–Alaoglu, Goldstine, and reflexivity

Theorem 2.5.1 (Banach–Alaoglu theorem)**.** The closed dual unit ball B_{E^*} is compact in the weak-$*$ topology.

Proof. Embed B_{E^*} into the product

$$K = \prod_{x \in E} \{z \in \mathbb{K} : |z| \le \|x\|\}$$

by $f \mapsto (f(x))_{x \in E}$. The product is compact by Tychonoff's theorem, and the product topology restricted to the image is exactly the weak-$*$ topology. The image is closed: linearity and the norm-one inequalities are expressed by closed coordinate equations

$$\omega_{x+y} = \omega_x + \omega_y, \qquad \omega_{\lambda x} = \lambda \omega_x, \qquad |\omega_x| \le \|x\|.$$

Therefore the image, and hence B_{E^*}, is compact. $\square$

Definition 2.5.2. The canonical embedding $J\colon E \to E^{**}$ is

$$(Jx)(f) = f(x).$$

The space E is *reflexive* if $J(E) = E^{**}$.

Theorem 2.5.3 (Goldstine theorem)**.** The canonical image $J(B_E)$ is weak-$*$ dense in $B_{E^{**}}$.

Proof. Let $x^{**} \in B_{E^{**}}$ and consider a basic weak-$*$ neighbourhood determined by $f_1, \dots, f_m \in E^*$. The finite-dimensional map

$$Q\colon E \to \mathbb{K}^m, \qquad Qx = (f_1(x), \dots, f_m(x)),$$

has the property that $Q(B_E)$ is dense enough to approximate $(x^{**}(f_1), \ldots, x^{**}(f_m))$. Otherwise finite-dimensional separation would yield scalars a_j with

$$\left| x^{**}\left(\sum_j a_j f_j\right) \right| > \left\| \sum_j a_j f_j \right\|,$$

contradicting $\|x^{**}\| \le 1$. Thus the neighbourhood meets $J(B_E)$. □

Theorem 2.5.4 (Kakutani's characterization)**.** A Banach space E is reflexive if and only if its closed unit ball is weakly compact.

Proof. If E is reflexive, J identifies B_E with $B_{E^{**}}$, which is weak-$*$ compact by Banach–Alaoglu; under J this is the weak topology of E. Conversely, if B_E is weakly compact, then $J(B_E)$ is weak-$*$ compact and hence weak-$*$ closed in E^{**}. Goldstine says it is weak-$*$ dense in $B_{E^{**}}$, so $J(B_E) = B_{E^{**}}$. Scaling gives $J(E) = E^{**}$. □

Theorem 2.5.5 (Eberlein–Šmulian theorem)**.** For a Banach space, a subset is weakly compact if and only if it is weakly sequentially compact.

Comment on the proof. This is a deep theorem: compactness and sequential compactness are not normally equivalent in nonmetrizable spaces. The source sketches the standard separable-reduction route, but not all of the required diagonal and closure arguments. We record the theorem in its correct form and use it below without presenting that sketch as a complete proof. □

Corollary 2.5.6. If E is reflexive, every closed bounded convex subset of E is weakly compact.

Proof. Such a set lies in a scalar multiple of B_E, which is weakly compact. It is weakly closed because convex norm-closed sets are weakly closed. □

2.6 Separability and metrizability

Theorem 2.6.1. If E^* is separable, then E is separable.

Proof. Choose a norm-dense sequence (f_n) in S_{E^*}. For each n, choose $x_n \in S_E$ with $|f_n(x_n)| > 1/2$. Let $M = \overline{\operatorname{span}\{x_n : n \ge 1\}}$. If $M \ne E$, Hahn–Banach gives $f \in S_{E^*}$ with $f|_M = 0$. Choose f_n with $\|f - f_n\| < 1/2$. Then

$$|f_n(x_n)| = |(f_n - f)(x_n)| < 1/2,$$

a contradiction. Thus $M = E$, and rational linear combinations of the x_n form a countable dense set. □

Theorem 2.6.2. For a Banach space E, the following are equivalent:

(a). E is separable;

(b). B_{E^*} is metrizable in the weak-$*$ topology.

Proof. If (x_n) is dense in B_E, then on B_{E^*} the metric

$$d(f,g) = \sum_{n=1}^{\infty} 2^{-n} \frac{|(f-g)(x_n)|}{1+|(f-g)(x_n)|}$$

induces the weak-$*$ topology. Approximation of an arbitrary $x \in E$ by multiples of the x_n, together with the uniform bound $\|f\|, \|g\| \leq 1$, proves the nontrivial inclusion of topologies.

Conversely, if B_{E^*} is weak-$*$ metrizable, Banach–Alaoglu makes it a compact metric space. Then $C(B_{E^*})$ is separable. The map $x \mapsto [f \mapsto f(x)]$ is an isometric embedding of E into $C(B_{E^*})$, so E is separable. □

Theorem 2.6.3. The weak topology on B_E is metrizable if and only if E^* is separable.

Proof. If (f_n) is norm dense in B_{E^*}, the analogous metric

$$d(x,y) = \sum_{n=1}^{\infty} 2^{-n} \frac{|f_n(x-y)|}{1+|f_n(x-y)|}$$

induces the weak topology on B_E.

Conversely, a countable neighbourhood base at 0 in B_E involves only a countable collection $\mathcal{F} \subseteq E^*$ of functionals. Let $M = \overline{\operatorname{span} \mathcal{F}}^{\|\cdot\|}$. If $M \neq E^*$, Hahn–Banach on E^* gives a nonzero x^{**} annihilating M. Goldstine then produces points of B_E that are arbitrarily small on every member of $\mathcal{F}$ but not on a fixed functional outside M, contradicting that the chosen countable family determines the weak topology. Hence $M = E^*$. □

Corollary 2.6.4. If E is separable, then every bounded sequence in E^* has a weak-$*$ convergent subsequence.

Proof. After scaling, the sequence lies in B_{E^*}, which is weak-$*$ compact and metrizable. Compact metric spaces are sequentially compact. □

2.7 Uniform convexity and reflexivity

Definition 2.7.1. A Banach space E is *uniformly convex* if, for every $\varepsilon > 0$, there is $\delta > 0$ such that

$$\|x\|, \|y\| \leq 1, \quad \|x-y\| \geq \varepsilon \quad \Longrightarrow \quad \left\| \frac{x+y}{2} \right\| \leq 1-\delta.$$

Proposition 2.7.2 (Radon–Riesz property)**.** Let E be uniformly convex. If $x_n \rightharpoonup x$ and $\|x_n\| \to \|x\|$, then $x_n \to x$ in norm.

Proof. The case $x = 0$ is immediate. Otherwise normalize $y_n = x_n/\|x_n\|$ and $y = x/\|x\|$. Then $y_n \rightharpoonup y$ and $\|y_n\| = \|y\| = 1$. If a subsequence satisfied $\|y_n - y\| \geq \varepsilon$, uniform convexity would give $\|(y_n + y)/2\| \leq 1-\delta$. But weak lower semicontinuity applied to $(y_n+y)/2 \rightharpoonup y$ gives $1 \leq \liminf \|(y_n+y)/2\|$, a contradiction. □

Theorem 2.7.3 (Milman–Pettis theorem)**.** Every uniformly convex Banach space is reflexive.

Proof. Let $x^{**} \in S_{E^{**}}$. By Goldstine there is a net $(x_\alpha) \subseteq B_E$ with $Jx_\alpha \xrightarrow{*} x^{**}$. Fix $\varepsilon > 0$ and choose the corresponding uniform-convexity constant $\delta > 0$. After multiplying a norming functional by a unimodular scalar, choose $f \in S_{E^*}$ with $\operatorname{Re} x^{**}(f) > 1 - \delta/2$. Eventually $\operatorname{Re} f(x_\alpha) > 1 - \delta$, and hence for large α, β,

$$\left\| \frac{x_\alpha + x_\beta}{2} \right\| \geq \operatorname{Re} f\left(\frac{x_\alpha + x_\beta}{2} \right) > 1 - \delta.$$

Uniform convexity gives $\|x_\alpha - x_\beta\| < \varepsilon$. Thus the net is norm Cauchy and converges to some $x \in B_E$. Since norm convergence implies weak-$*$ convergence under J, uniqueness of the weak-$*$ limit gives $Jx = x^{**}$. Scaling proves that J is surjective. □

Chapter 3 Lebesgue Spaces

Throughout this chapter, $\Omega \subseteq \mathbb{R}^d$ is measurable and all L^p spaces are taken with respect to Lebesgue measure unless stated otherwise. For $1 < p < \infty$, the conjugate exponent is $p' = p/(p-1)$.

3.1 Uniform convexity, reflexivity, and duality

Theorem 3.1.1. For $1 < p < \infty$, the space $L^p(\Omega)$ is uniformly convex and therefore reflexive.

Proof. Clarkson's inequalities give, for $2 \le p < \infty$,

$$\left\| \frac{f+g}{2} \right\|_p^p + \left\| \frac{f-g}{2} \right\|_p^p \le \frac{\|f\|_p^p + \|g\|_p^p}{2},$$

and the dual Clarkson inequality gives the corresponding estimate for $1 < p \le 2$. If $\|f\|_p, \|g\|_p \le 1$ and $\|f-g\|_p \ge \varepsilon$, these inequalities bound $\|(f+g)/2\|_p$ by $1 - \delta_p(\varepsilon)$ for some positive $\delta_p(\varepsilon)$. Thus L^p is uniformly convex, and the Milman–Pettis theorem yields reflexivity. □

Theorem 3.1.2 (Riesz representation for L^p)**.** Let $1 < p < \infty$. For each $g \in L^{p'}(\Omega)$, the formula

$$T_g(f) = \int_\Omega fg \, \mathrm{d}x$$

defines an element of $(L^p(\Omega))^*$, and

$$\|T_g\| = \|g\|_{p'}.$$

Every element of $(L^p(\Omega))^*$ is of this form. Hence

$$(L^p(\Omega))^* \cong L^{p'}(\Omega)$$

isometrically.

Proof. Hölder's inequality gives $\|T_g\| \le \|g\|_{p'}$. Taking

$$f = \frac{|g|^{p'-2}\overline{g}}{\|g\|_{p'}^{p'-1}}$$

(with the evident real-valued version) gives equality when $g \neq 0$.

Let $J\colon L^{p'} \to (L^p)^*$ be $Jg = T_g$. It is an isometry, so its range is closed. If $\Phi \in (L^p)^{**}$ annihilates $J(L^{p'})$, reflexivity of L^p gives $\Phi = J_{L^p} f$ for some $f \in L^p$, and then

$$0 = \Phi(Jg) = \int_\Omega fg \, \mathrm{d}x \qquad (g \in L^{p'}).$$

Choosing the usual truncations of $|f|^{p-2}\overline{f}$ gives $f = 0$. Therefore $J(L^{p'})$ has zero annihilator and is weakly dense; since it is a closed linear subspace, it equals $(L^p)^*$. □

Theorem 3.1.3 (The endpoint duality theorem)**.** For a σ-finite measure space, and in particular for Lebesgue measure on $\Omega \subseteq \mathbb{R}^d$,

$$(L^1(\Omega))^* \cong L^\infty(\Omega)$$

isometrically through the same pairing $T_g(f) = \int_\Omega fg$.

Proof sketch. Hölder's inequality gives the isometric embedding $L^\infty \hookrightarrow (L^1)^*$. Given $T \in (L^1)^*$, define the signed or complex measure $\nu(A) = T(\mathbf{1}_A)$ on sets of finite measure. It is absolutely continuous with respect to Lebesgue measure and satisfies $|\nu(A)| \leq \|T\|\,|A|$. Radon–Nikodym gives a density $g \in L^\infty$ with $\|g\|_\infty \leq \|T\|$, and density of simple functions then yields $T(f) = \int fg$. The reverse norm inequality follows by testing on sets where $|g|$ is close to its essential supremum. □

Remark 3.1.4. Except in the finite-dimensional case (equivalently, when the measure algebra has only finitely many atoms), $L^1(\Omega)$ is not reflexive. Indeed, if it were reflexive then its dual $L^\infty(\Omega)$ would also be reflexive, which fails on every genuinely infinite measure algebra. Likewise $L^\infty(\Omega)$ is nonreflexive except in the same finite-dimensional situation.

3.2 Separability and dense classes

Theorem 3.2.1. For $1 \leq p < \infty$, the space $L^p(\mathbb{R}^d)$ is separable. The space $L^\infty(\mathbb{R}^d)$ is not separable.

Proof. Finite linear combinations, with rational coefficients, of indicators of rectangles with rational endpoints form a countable set. Simple-function approximation and regularity of Lebesgue measure show that this set is dense in L^p for $p < \infty$. For L^∞, choose infinitely many pairwise measurably distinguishable sets A_α such that $\|\mathbf{1}_{A_\alpha} - \mathbf{1}_{A_\beta}\|_\infty = 1$ whenever $\alpha \neq \beta$; there are uncountably many such indicators, so no countable set can be dense. □

Theorem 3.2.2. For $1 \leq p < \infty$, $C_c(\mathbb{R}^d)$ and $C_c^\infty(\mathbb{R}^d)$ are dense in $L^p(\mathbb{R}^d)$.

Proof. First approximate by compactly supported simple functions, then use regularity of Lebesgue measure and Urysohn-type cutoff functions to approximate their indicators by functions in C_c. Mollification of a compactly supported continuous function gives C_c^∞ functions converging uniformly, hence in L^p. □

3.3 Convolution and mollifiers

Theorem 3.3.1 (Young's convolution inequality). Let $1 \le p \le \infty$, $f \in L^1(\mathbb{R}^d)$, and $g \in L^p(\mathbb{R}^d)$. Then

$$(f * g)(x) = \int_{\mathbb{R}^d} f(x-y)g(y)\,\mathrm{d}y$$

is defined for almost every x, belongs to L^p, and satisfies

$$\|f * g\|_p \le \|f\|_1 \|g\|_p.$$

Proof. For $p = 1$ and $p = \infty$, use Tonelli's theorem and the essential-supremum bound. For $1 < p < \infty$, Hölder's inequality with respect to the measure $|f(x-y)|\,\mathrm{d}y$ gives

$$|(f * g)(x)|^p \le \|f\|_1^{p-1} \int |f(x-y)|\,|g(y)|^p\,\mathrm{d}y.$$

Integrating in x and applying Tonelli yields the result. □

Proposition 3.3.2 (Differentiation of convolution). If $f \in C_c^k(\mathbb{R}^d)$ and $g \in L^1_{\mathrm{loc}}(\mathbb{R}^d)$, then $f * g \in C^k$ wherever the convolution is defined, and

$$D^\alpha (f * g) = (D^\alpha f) * g \qquad (|\alpha| \le k).$$

Definition 3.3.3. A sequence (ρ_n) is an *approximate identity* or a sequence of *mollifiers* if

$$\rho_n \in C_c^\infty(\mathbb{R}^d), \qquad \rho_n \ge 0, \qquad \int \rho_n = 1, \qquad \operatorname{supp} \rho_n \subseteq B(0, 1/n).$$

A standard choice is $\rho_n(x) = n^d \rho(nx)$ for a fixed nonnegative $\rho \in C_c^\infty(B(0,1))$ with integral one.

Proposition 3.3.4. Let (ρ_n) be mollifiers.

(a). If $f \in C_c(\mathbb{R}^d)$, then $\rho_n * f \to f$ uniformly.

(b). If $f \in L^p(\mathbb{R}^d)$ with $1 \le p < \infty$, then $\rho_n * f \to f$ in L^p.

Proof. For continuous compactly supported f,

$$|(\rho_n * f)(x) - f(x)| \le \int \rho_n(y)|f(x-y) - f(x)|\,\mathrm{d}y,$$

and uniform continuity makes the right-hand side tend uniformly to zero. The L^p statement follows by approximating f by a function in C_c, using Young's inequality for the two error terms. □

Corollary 3.3.5 (Vanishing weak gradient). Let $\Omega \subseteq \mathbb{R}^d$ be open and connected, and let $u \in L^1_{\mathrm{loc}}(\Omega)$. If

$$\int_\Omega u\,\partial_j \varphi\,\mathrm{d}x = 0 \qquad (\varphi \in C_c^\infty(\Omega),\ 1 \le j \le d),$$

then u is almost everywhere equal to a constant on Ω.

Proof. On every ball B compactly contained in Ω, mollification gives smooth functions u_ε with $\nabla u_\varepsilon = 0$ on a slightly smaller ball. Hence each u_ε is constant there. Passing to the L^1_{loc} limit shows that u is almost everywhere constant on that ball. Overlapping balls propagate the same constant throughout each connected component; connectedness gives a single constant on Ω. □

Chapter 4 Hilbert Spaces

4.1 Orthogonality and direct sums

Proposition 4.1.1 (Orthogonal density criterion)**.** Let H be a Hilbert space and let $M \subseteq H$ be a linear subspace. Then

$$\overline{M} = H \iff M^{\perp} = \{0\}.$$

Proof. If $\overline{M} = H$, continuity of the inner product makes every vector orthogonal to M orthogonal to all of H, hence zero. Conversely, if $\overline{M} \neq H$, the projection theorem gives a nonzero component of some $x \notin \overline{M}$ in $(\overline{M})^{\perp} = M^{\perp}$. □

Proposition 4.1.2. For any nonempty subset S of an inner-product space,

$$S^{\perp} = \{x : \langle x, s\rangle = 0 \text{ for every } s \in S\}$$

is a closed linear subspace. In a Hilbert space,

$$S^{\perp} = (\overline{\operatorname{span} S})^{\perp}, \qquad S^{\perp\perp} = \overline{\operatorname{span} S}.$$

Moreover, $S \cap S^{\perp} = \{0\}$ whenever S is a subspace.

Theorem 4.1.3 (Orthogonal direct-sum theorem)**.** Let M be a closed linear subspace of a Hilbert space H. Then

$$H = M \oplus M^{\perp}.$$

Thus every $x \in H$ has a unique decomposition

$$x = P_M x + (x - P_M x), \qquad P_M x \in M, \quad x - P_M x \in M^{\perp}.$$

4.2 Inner products and projections

Definition 4.2.1. An inner product on a real vector space is a symmetric positive-definite bilinear form. On a complex vector space it is a positive-definite sesquilinear form, conjugate linear in one variable and linear in the other. The induced norm is

$$\|x\| = \sqrt{\langle x, x\rangle}.$$

A complete inner-product space is a *Hilbert space*.

Proposition 4.2.2. Every inner product satisfies the Cauchy–Schwarz inequality

$$|\langle x, y\rangle| \leq \|x\|\,\|y\|$$

and the parallelogram identity

$$\|x+y\|^2 + \|x-y\|^2 = 2\|x\|^2 + 2\|y\|^2.$$

Conversely, a norm comes from an inner product if and only if it satisfies the parallelogram identity; the inner product is recovered by polarization.

Theorem 4.2.3 (Projection theorem)**.** Let K be a nonempty closed convex subset of a Hilbert space H. For every $f \in H$ there is a unique $u \in K$ satisfying

$$\|f-u\| = \operatorname{dist}(f, K).$$

It is characterized by the variational inequality

$$\operatorname{Re}\langle f-u, v-u\rangle \leq 0 \qquad (v \in K).$$

We write $u = P_K f$.

Proof. Let $d = \operatorname{dist}(f, K)$ and choose a minimizing sequence $(u_n) \subseteq K$. Convexity gives $(u_n + u_m)/2 \in K$. The parallelogram identity yields

$$\|u_n - u_m\|^2 = 2\|f-u_n\|^2 + 2\|f-u_m\|^2 - 4\left\|f - \frac{u_n+u_m}{2}\right\|^2,$$

which tends to zero. Thus (u_n) is Cauchy and converges to some $u \in K$; continuity gives $\|f-u\| = d$. For $v \in K$ and $0 < t \leq 1$, the point $u + t(v-u)$ lies in K. Minimality implies

$$\|f-u\|^2 \leq \|f-u-t(v-u)\|^2 = \|f-u\|^2 - 2t\operatorname{Re}\langle f-u, v-u\rangle + t^2\|v-u\|^2.$$

Divide by t and let $t \downarrow 0$ to obtain the inequality. Conversely, the inequality gives, for $v \in K$,

$$\|f-v\|^2 = \|f-u\|^2 + \|u-v\|^2 + 2\operatorname{Re}\langle f-u, u-v\rangle \geq \|f-u\|^2.$$

If u_1, u_2 both satisfy the variational inequalities, use $v = u_2$ in the first and $v = u_1$ in the second; adding gives $\|u_1 - u_2\|^2 \leq 0$. □

Proposition 4.2.4 (Nonexpansiveness of metric projection)**.** For every nonempty closed convex $K \subseteq H$,

$$\|P_K f - P_K g\| \leq \|f-g\| \qquad (f, g \in H).$$

Proof. Apply the variational inequality to $P_K f$ with competitor $P_K g$ and to $P_K g$ with competitor $P_K f$.

Adding gives

$$\|P_K f - P_K g\|^2 \le \operatorname{Re}\langle P_K f - P_K g, f - g\rangle \le \|P_K f - P_K g\|\,\|f - g\|.$$

□

Corollary 4.2.5 (Orthogonal projections). If M is a closed linear subspace, then P_M is linear, bounded, and $\|P_M\| \le 1$. It is characterized by

$$P_M f \in M, \qquad f - P_M f \in M^\perp.$$

If $M \ne \{0\}$, then $\|P_M\| = 1$.

Proof. The variational inequality may be tested with $u \pm v$ and, in the complex case, $u \pm iv$ for $v \in M$, so it reduces to orthogonality. Uniqueness of the orthogonal decomposition gives linearity. Nonexpansiveness gives the norm bound. □

4.3 Riesz representation

Theorem 4.3.1 (Riesz representation theorem). For every $\ell \in H^*$ there is a unique $y \in H$ such that

$$\ell(x) = \langle x, y\rangle \qquad (x \in H)$$

(with the variables adjusted to the chosen complex convention), and $\|\ell\| = \|y\|$.

Proof. If $\ell = 0$, take $y = 0$. Otherwise $M = \operatorname{Ker}\ell$ is a closed proper subspace. Choose $0 \ne z \in M^\perp$. For each $x \in H$, the vector

$$x - \frac{\ell(x)}{\ell(z)} z$$

belongs to M, hence is orthogonal to z. Therefore

$$\ell(x) = \frac{\ell(z)}{\|z\|^2}\langle x, z\rangle,$$

which has the required form. Cauchy–Schwarz gives $\|\ell\| \le \|y\|$, while testing at $x = y/\|y\|$ gives equality. □

4.4 Orthogonal sums and orthonormal bases

Definition 4.4.1. Let $(E_n)_{n\ge1}$ be closed subspaces of a Hilbert space H. Their Hilbert direct sum is

$$\bigoplus_{n=1}^{\infty} E_n = \left\{ (u_n) : u_n \in E_n,\ \sum_{n=1}^{\infty} \|u_n\|^2 < \infty \right\}$$

with inner product $\langle (u_n), (v_n)\rangle = \sum_n \langle u_n, v_n\rangle$.

Lemma 4.4.2 (Orthogonal series). If (v_n) is an orthogonal sequence in H and $\sum_n \|v_n\|^2 < \infty$, then $\sum_n v_n$

converges in H and

$$\left\| \sum_{n=1}^{\infty} v_n \right\|^2 = \sum_{n=1}^{\infty} \|v_n\|^2.$$

Proof. For $m > n$, orthogonality gives

$$\left\| \sum_{k=n+1}^{m} v_k \right\|^2 = \sum_{k=n+1}^{m} \|v_k\|^2,$$

so the partial sums are Cauchy. The norm identity follows by passage to the limit. □

Theorem 4.4.3 (Projections onto mutually orthogonal subspaces)**.** Let (E_n) be mutually orthogonal closed subspaces and let $P_n = P_{E_n}$. For all $u \in H$,

$$\sum_{n=1}^{\infty} \|P_n u\|^2 \le \|u\|^2,$$

and the series $\sum_n P_n u$ converges. If $E = \overline{\operatorname{span} \bigcup_n E_n}$, then

$$u = P_E u = \sum_{n=1}^{\infty} P_n u, \qquad \|P_E u\|^2 = \sum_{n=1}^{\infty} \|P_n u\|^2.$$

Proof. Finite orthogonality gives

$$\left\| \sum_{n=1}^{N} P_n u \right\|^2 = \sum_{n=1}^{N} \|P_n u\|^2 = \left\langle u, \sum_{n=1}^{N} P_n u \right\rangle \le \|u\| \left\| \sum_{n=1}^{N} P_n u \right\|.$$

This yields Bessel's bound and convergence. The limit lies in E, and the difference from u is orthogonal to every E_n, hence to E; it is therefore $P_E u$. □

Definition 4.4.4. A sequence (e_n) is an *orthonormal basis* of H if

$$\langle e_n, e_m \rangle = \delta_{nm} \qquad \text{and} \qquad \overline{\operatorname{span}\{e_n : n \ge 1\}} = H.$$

Theorem 4.4.5 (Fourier expansion and Parseval identity)**.** If (e_n) is an orthonormal basis, then for every $u \in H$,

$$u = \sum_{n=1}^{\infty} \langle u, e_n \rangle e_n, \qquad \|u\|^2 = \sum_{n=1}^{\infty} |\langle u, e_n \rangle|^2.$$

Conversely, for every $(a_n) \in \ell^2$ the series $\sum_n a_n e_n$ converges to a unique $u \in H$ satisfying

$$\langle u, e_n \rangle = a_n, \qquad \|u\|^2 = \sum_n |a_n|^2.$$

Theorem 4.4.6. Every separable Hilbert space has a countable orthonormal basis.

Proof. Choose a countable dense set, discard vectors already in the span of their predecessors, and apply the Gram–Schmidt process. The resulting orthonormal sequence has dense linear span because it spans the same finite-dimensional spaces at each stage as the retained dense sequence. □

Chapter 5 Spectral Theory

Throughout this chapter, scalar fields are complex whenever the spectrum of an operator is discussed.

5.1 Closed operators, resolvents, and spectra

Definition 5.1.1. Let X be a complex Banach space and let $T\colon \mathrm{D}(T) \subseteq X \to X$ be linear. A number $\lambda \in \mathbb{C}$ is a *regular value* if

$$T - \lambda I\colon \mathrm{D}(T) \to X$$

is bijective and its inverse

$$R_\lambda(T) = (T - \lambda I)^{-1}\colon X \to X$$

is bounded. The *resolvent set* $\rho(T)$ is the set of regular values, and the *spectrum* is

$$\sigma(T) = \mathbb{C} \setminus \rho(T).$$

The spectrum is partitioned into:

- the point spectrum, where $T - \lambda I$ is not injective;
- the continuous spectrum, where it is injective and has dense but non-surjective range;
- the residual spectrum, where it is injective but its range is not dense.

Proposition 5.1.2. If T is closed and $T - \lambda I$ is bijective, then $R_\lambda(T)$ is automatically bounded. In particular, for a closed operator the definition of $\rho(T)$ may omit boundedness of the inverse.

Proof. The inverse has graph obtained from $\mathrm{Graph}(T - \lambda I)$ by interchanging the two coordinates, so it is closed. Its domain is all of the Banach space X; the closed graph theorem makes it bounded. □

Theorem 5.1.3 (Neumann series)**.** Let $A \in \mathcal{B}(X, X)$ with $\|A\| < 1$. Then $I - A$ is invertible and

$$(I - A)^{-1} = \sum_{n=0}^{\infty} A^n$$

in operator norm. More generally, if $\lambda_0 \in \rho(T)$ and $|\lambda - \lambda_0|\, \|R_{\lambda_0}(T)\| < 1$, then $\lambda \in \rho(T)$ and

$$R_\lambda(T) = R_{\lambda_0}(T) \sum_{n=0}^{\infty} \big((\lambda - \lambda_0) R_{\lambda_0}(T)\big)^n.$$

Proof. The partial sums telescope: $(I - A)\sum_{n=0}^{N} A^n = I - A^{N+1}$, and $A^{N+1} \to 0$. For the resolvent, factor

$$T - \lambda I = \big(I - (\lambda - \lambda_0)R_{\lambda_0}(T)\big)(T - \lambda_0 I).$$

$\square$

Corollary 5.1.4. The resolvent set is open and the spectrum is closed. If $T \in \mathcal{B}(X, X)$, then

$$\sigma(T) \subseteq \{\lambda : |\lambda| \le \|T\|\}.$$

Proof. Openness follows from the local Neumann-series formula. If $|\lambda| > \|T\|$, then

$$T - \lambda I = -\lambda \left(I - \frac{T}{\lambda}\right)$$

is invertible by the Neumann series. $\square$

Theorem 5.1.5 (Resolvent identity)**.** For $\lambda, \mu \in \rho(T)$,

$$R_\lambda(T) - R_\mu(T) = (\lambda - \mu)R_\lambda(T)R_\mu(T).$$

In particular, the resolvents commute.

Proof. Use $A^{-1} - B^{-1} = A^{-1}(B - A)B^{-1}$ with $A = T - \lambda I$ and $B = T - \mu I$. $\square$

Theorem 5.1.6. For every bounded operator on a nonzero complex Banach space, the spectrum is nonempty and compact.

Proof. Closedness and boundedness were proved above. If $\sigma(T) = \varnothing$, then for each $x \in X$ and $f \in X^*$ the scalar function

$$\lambda \longmapsto f(R_\lambda(T)x)$$

is entire. For $|\lambda| > \|T\|$, the Neumann expansion gives $\|R_\lambda(T)\| \le (|\lambda| - \|T\|)^{-1}$, so this entire function tends to zero at infinity. Liouville's theorem makes it identically zero. Since functionals separate points, $R_\lambda(T) = 0$, impossible for an inverse. $\square$

Definition 5.1.7. For $T \in \mathcal{B}(X, X)$, the *spectral radius* is

$$r(T) = \sup\{|\lambda| : \lambda \in \sigma(T)\}.$$

Proposition 5.1.8. If $\lambda \in \rho(T)$, then $R_\lambda(T)$ commutes with T. More generally, $R_\lambda(T)$ commutes with every bounded operator that commutes with T. Consequently, all resolvents commute with one another.

5.2 Spectral mapping and the spectral radius formula

Theorem 5.2.1 (Polynomial spectral mapping theorem)**.** Let $T \in \mathcal{B}(X, X)$ and let p be a complex polynomial. Then

$$\sigma(p(T)) = p(\sigma(T)).$$

Proof. First let $\mu \notin p(\sigma(T))$. Factor

$$p(z) - \mu = c \prod_{j=1}^{m} (z - \lambda_j).$$

Every λ_j lies in $\rho(T)$, so $p(T) - \mu I = c \prod_j (T - \lambda_j I)$ is invertible.

Conversely, let $\mu = p(\lambda)$ with $\lambda \in \sigma(T)$. Polynomial division gives

$$p(T) - p(\lambda)I = (T - \lambda I)q(T),$$

and the two factors commute. If the left side were invertible, then $(T - \lambda I)$ would have both a bounded left inverse and a bounded right inverse obtained by combining $q(T)$ with that inverse, hence would itself be invertible. This contradicts $\lambda \in \sigma(T)$. □

Proposition 5.2.2 (Linear independence of eigenvectors)**.** Eigenvectors corresponding to distinct eigenvalues are linearly independent.

Proof. If $\sum_{j=1}^{n} a_j x_j = 0$ is a shortest nontrivial relation and $Tx_j = \lambda_j x_j$, apply $T - \lambda_n I$. The resulting relation among $x_1, \ldots, x_{n-1}$ has coefficients $a_j(\lambda_j - \lambda_n)$ and is nontrivial, a contradiction. □

Theorem 5.2.3 (Spectral radius formula)**.** For $T \in \mathcal{B}(X, X)$,

$$r(T) = \lim_{n \to \infty} \|T^n\|^{1/n} = \inf_{n \geq 1} \|T^n\|^{1/n}.$$

Proof. By spectral mapping, $r(T)^n = r(T^n) \leq \|T^n\|$, so $r(T) \leq \inf_n \|T^n\|^{1/n}$.

Let $R > r(T)$. The resolvent is analytic on $|\lambda| > r(T)$ and has the Laurent expansion at infinity

$$R_\lambda(T) = -\sum_{n=0}^{\infty} \frac{T^n}{\lambda^{n+1}}$$

for $|\lambda| > \|T\|$. Cauchy's formula, applied on the circle $|\lambda| = R$, gives

$$T^n = -\frac{1}{2\pi i} \int_{|\lambda|=R} \lambda^n R_\lambda(T)\, \mathrm{d}\lambda.$$

Hence $\|T^n\| \leq C_R R^{n+1}$, and $\limsup_n \|T^n\|^{1/n} \leq R$. Letting $R \downarrow r(T)$ gives the reverse inequality. The equality with the infimum follows from submultiplicativity of $\|T^n\|$ and Fekete's lemma applied to $\log \|T^n\|$. □

5.3 Compact operators

Definition 5.3.1. A bounded linear operator $T\colon X \to Y$ is compact if it maps bounded sets to relatively compact sets. Equivalently, for every bounded sequence (x_n) in X, the sequence (Tx_n) has a convergent subsequence in Y.

Lemma 5.3.2. Every compact linear operator is bounded. The identity operator on a normed space is compact if and only if the space is finite dimensional.

Proof. If T were unbounded, choose x_n with $\|x_n\| \le 1$ and $\|Tx_n\| \ge n$; then (Tx_n) has no convergent subsequence. The statement about the identity is the Riesz lemma characterization of finite-dimensional normed spaces. □

Theorem 5.3.3 (Sequential criterion)**.** A linear map $T\colon X \to Y$ is compact if and only if every bounded sequence (x_n) has a subsequence (x_{n_k}) for which (Tx_{n_k}) converges.

Theorem 5.3.4. Every finite-rank operator is compact. If Y is Banach, then $\mathcal{K}(X,Y)$ is closed in $\mathcal{B}(X,Y)$ for the operator norm.

Proof. A bounded set in a finite-dimensional range is relatively compact. For the closure statement, let $T_n \to T$ in operator norm with each T_n compact. Given a bounded sequence, successively choose subsequences on which $T_1, T_2, \ldots$ converge and take the diagonal subsequence. Uniform operator-norm approximation then makes the image under T Cauchy. Completeness of Y gives convergence. □

Definition 5.3.5. A metric-space subset B is *totally bounded* if, for every $\varepsilon > 0$, it is covered by finitely many balls of radius ε.

Lemma 5.3.6. A metric space is compact if and only if it is complete and totally bounded. Every totally bounded set is separable, and every subset of a totally bounded set is totally bounded.

Theorem 5.3.7. If $T\colon X \to Y$ is compact, then $\operatorname{Ran}(T)$ is separable.

Proof. Write

$$\operatorname{Ran}(T) = \bigcup_{n=1}^{\infty} T(nB_X).$$

Each $T(nB_X)$ is totally bounded and therefore separable. A countable union of countable dense sets is dense in the range. □

Theorem 5.3.8 (Compact extension from a dense subspace)**.** Let X and Y be Banach spaces, let $D \subseteq X$ be dense, and let $T\colon D \to Y$ be bounded. Its unique bounded extension $\widetilde{T}\colon X \to Y$ is compact if T maps the unit ball of D to a totally bounded subset of Y.

Proof. For a bounded sequence $(x_n) \subset X$, approximate each x_n by $d_n \in D$ so closely that $\|x_n - d_n\| \to 0$, while keeping (d_n) bounded. Total boundedness gives a subsequence for which Td_n is Cauchy. Since

$\|\widetilde{T}x_n - Td_n\| \to 0$, the corresponding $\widetilde{T}x_n$ is Cauchy and converges in Y. □

Theorem 5.3.9 (Schauder theorem)**.** A bounded linear operator $T\colon X \to Y$ is compact if and only if $T^*\colon Y^* \to X^*$ is compact.

Proof. Assume T is compact. To show that $T^*(B_{Y^*})$ is totally bounded, fix $\varepsilon > 0$ and choose a finite ε-net $Tx_1, \dots, Tx_m$ for $T(B_X)$. Define

$$A\colon Y^* \to \mathbb{K}^m, \qquad A(y^*) = (y^*(Tx_1), \dots, y^*(Tx_m)).$$

The bounded set $A(B_{Y^*})$ has a finite net. If y^*, z^* have close A-coordinates, then for each $x \in B_X$, choosing x_j with $\|Tx - Tx_j\| < \varepsilon$ gives

$$|(T^*y^* - T^*z^*)(x)| \le |(y^* - z^*)(Tx_j)| + 2\varepsilon.$$

Thus a finite net in the coordinate image yields a finite multiple of ε-net for $T^*(B_{Y^*})$.
Applying the proved implication to T^* shows that T^{**} is compact. The canonical embeddings satisfy $T^{**}J_X = J_Y T$, and J_X, J_Y are isometries. Hence T is compact. □

Lemma 5.3.10 (Ideal property)**.** For normed spaces X, Y, compact operators form a two-sided operator ideal: if $T \in \mathcal{K}(X, Y)$ and A, B are bounded with compatible domains, then ATB is compact.

5.4 The spectrum of a compact operator

Theorem 5.4.1 (Riesz spectral theorem for compact operators)**.** Let T be a compact operator on an infinite-dimensional complex Banach space. Then:

(a). every nonzero point of $\sigma(T)$ is an eigenvalue;

(b). for each $\lambda \neq 0$, the eigenspace $\mathrm{Ker}(T - \lambda I)$ is finite dimensional;

(c). $\sigma(T)$ is at most countable, and 0 is its only possible accumulation point.

Proof. For finite dimensionality, the identity on $N_\lambda = \mathrm{Ker}(T - \lambda I)$ equals $\lambda^{-1}T|_{N_\lambda}$ and is compact, so N_λ is finite dimensional.
Fix $\varepsilon > 0$. If there were infinitely many distinct eigenvalues $|\lambda_n| \ge \varepsilon$, choose unit eigenvectors x_n inductively so that x_n has distance at least $1/2$ from $M_{n-1} = \mathrm{span}\{x_1, \dots, x_{n-1}\}$. Since M_{n-1} is invariant under T, for $m < n$ one obtains

$$\|Tx_n - Tx_m\| = \|\lambda_n x_n - \lambda_m x_m\| \ge c\varepsilon$$

for an absolute $c > 0$, contradicting compactness. Hence only finitely many eigenvalues lie outside every disk about 0, proving countability and the accumulation assertion.
The remaining claim—that every nonzero spectral value is an eigenvalue—is a consequence of the Fredholm alternative proved below. □

Proposition 5.4.2. Let T be compact and $\lambda \neq 0$. Then $\mathrm{Ran}(T - \lambda I)$ is closed.

Proof. Let $N = \operatorname{Ker}(T - \lambda I)$, which is finite dimensional, and choose a closed complement $X = N \oplus M$. The restriction $S = (T - \lambda I)|_M$ is bounded below. Otherwise there are $x_n \in M$ with $\|x_n\| = 1$ and $Sx_n \to 0$. Compactness gives a subsequence for which Tx_n converges, and

$$x_n = \lambda^{-1}(Tx_n - Sx_n)$$

then converges to some $x \in M$ with $\|x\| = 1$ and $Sx = 0$, contradicting $M \cap N = \{0\}$. A bounded-below operator has closed range, and $\operatorname{Ran} S = \operatorname{Ran}(T - \lambda I)$. $\square$

Theorem 5.4.3 (Stabilization and Riesz decomposition)**.** Let T be compact and $\lambda \neq 0$. There is $m \geq 1$ such that

$$\operatorname{Ker}(T - \lambda I)^m = \operatorname{Ker}(T - \lambda I)^{m+1} = \cdots,$$

$$\operatorname{Ran}(T - \lambda I)^m = \operatorname{Ran}(T - \lambda I)^{m+1} = \cdots,$$

and

$$X = \operatorname{Ker}(T - \lambda I)^m \oplus \operatorname{Ran}(T - \lambda I)^m.$$

Proof sketch. If the kernels increased strictly forever, Riesz's lemma would supply unit vectors x_n in successive kernels separated from their predecessors. The identity

$$Tx_n = \lambda x_n + (T - \lambda I)x_n$$

and invariance of the earlier kernels would then make (Tx_n) uniformly separated, contradicting compactness. Applying the same argument to the adjoint, together with annihilator identities, yields stabilization of the ranges. At the first common stabilization index, the intersection of the kernel and range is zero and the quotient dimensions show that their sum is all of X. $\square$

5.5 Fredholm alternative

Lemma 5.5.1 (Biorthogonal vectors)**.** If $f_1, \ldots, f_n \in X^*$ are linearly independent, then there exist $x_1, \ldots, x_n \in X$ such that

$$f_i(x_j) = \delta_{ij}.$$

Proof. The map $Q\colon X \to \mathbb{K}^n$, $Qx = (f_1(x), \ldots, f_n(x))$, is surjective; otherwise a nonzero functional on $\mathbb{K}^n$ annihilating $Q(X)$ would give a nontrivial linear relation among the f_i. Choose x_j with $Qx_j = e_j$. $\square$

Theorem 5.5.2 (Fredholm alternative)**.** Let $T \in \mathcal{K}(X, X)$ and let $\lambda \neq 0$. Put $S = \lambda I - T$. Then:

(a). S is injective if and only if it is surjective; in this case S^{-1} is bounded;

(b). $\dim \operatorname{Ker} S = \dim \operatorname{Ker} S^* < \infty$;

(c). the equation $Sx = y$ is solvable if and only if

$$f(y) = 0 \qquad (f \in \operatorname{Ker} S^*);$$

(d). the equation $S^* f = g$ is solvable if and only if

$$g(x) = 0 \qquad (x \in \operatorname{Ker} S).$$

Here $S^* = \lambda I - T^*$ for the Banach-space adjoint defined on linear functionals. Under the Hilbert-space adjoint identification, the corresponding spectral parameter is conjugated according to the inner-product convention.

Proof. The range of S is closed. If S is injective but not surjective, the strictly decreasing closed ranges $X \supsetneq \operatorname{Ran} S \supsetneq \operatorname{Ran} S^2 \supsetneq \cdots$ would contradict the range stabilization theorem. If S is surjective but not injective, the strictly increasing kernels $\operatorname{Ker} S \subsetneq \operatorname{Ker} S^2 \subsetneq \cdots$ would contradict kernel stabilization. Thus injectivity and surjectivity are equivalent, and the bounded inverse theorem gives boundedness of the inverse.

The annihilator identity

$$\overline{\operatorname{Ran} S} = {}^{\perp}\operatorname{Ker} S^*$$

and closedness of $\operatorname{Ran} S$ give the solvability criterion for $Sx = y$. Applying the same result to S^* and using Schauder's theorem gives the dual criterion. Finally, biorthogonal vectors and the two solvability criteria show that neither $\operatorname{Ker} S$ nor $\operatorname{Ker} S^*$ can have larger dimension than the other; hence their finite dimensions are equal. □

Corollary 5.5.3. Every nonzero spectral value of a compact operator is an eigenvalue.

Proof. If $\lambda \neq 0$ and $\lambda I - T$ were injective, the Fredholm alternative would make it surjective and hence invertible. Thus a nonzero point of the spectrum must have nontrivial kernel. □

5.6 Arzelà–Ascoli and integral operators

Theorem 5.6.1 (Arzelà–Ascoli theorem)**.** Let K be a compact metric space. A bounded equicontinuous family $\mathcal{F} \subseteq C(K)$ is relatively compact in the supremum norm. Equivalently, every sequence in $\mathcal{F}$ has a uniformly convergent subsequence.

Theorem 5.6.2 (Continuous-kernel integral operators)**.** Let $J = [a, b]$ and let $k \in C(J \times J)$. Define

$$(Tf)(s) = \int_a^b k(s,t) f(t)\, \mathrm{d}t, \qquad f \in C(J).$$

Then $T\colon C(J) \to C(J)$ is compact.

Proof. For $\|f\|_\infty \le 1$,

$$\|Tf\|_\infty \le (b-a)\|k\|_\infty,$$

so the image of the unit ball is uniformly bounded. Uniform continuity of k gives

$$|Tf(s_1) - Tf(s_2)| \le \int_a^b |k(s_1,t) - k(s_2,t)|\, \mathrm{d}t,$$

which tends to zero uniformly in f as $s_1 \to s_2$. Thus the image is equicontinuous, and Arzelà–Ascoli gives relative compactness. □

Chapter 6 Operator Semigroups and Hille–Yosida

6.1 Strongly continuous semigroups and generators

Definition 6.1.1. Let X be a Banach space. A family $(T(t))_{t\geq 0} \subseteq \mathcal{B}(X,X)$ is a *strongly continuous semigroup*, or C_0-semigroup, if

$$T(0) = I, \qquad T(t+s) = T(t)T(s),$$

and

$$\lim_{t\downarrow 0} T(t)x = x \qquad (x \in X).$$

Its *infinitesimal generator* is

$$Ax = \lim_{t\downarrow 0} \frac{T(t)x - x}{t},$$

with domain

$$\mathrm{D}(A) = \left\{ x \in X : \lim_{t\downarrow 0} \frac{T(t)x - x}{t} \text{ exists in } X \right\}.$$

Proposition 6.1.2. Let A be the generator of a C_0-semigroup $(T(t))$. Then:

(a). A is linear and $\mathrm{D}(A)$ is dense in X;

(b). $T(t)\,\mathrm{D}(A) \subseteq \mathrm{D}(A)$ and $AT(t)x = T(t)Ax$ for $x \in \mathrm{D}(A)$;

(c). for $x \in \mathrm{D}(A)$, the orbit is continuously differentiable and

$$\frac{\mathrm{d}}{\mathrm{d}t} T(t)x = AT(t)x = T(t)Ax;$$

(d). A is closed.

Proof. Linearity follows from the limit. For density, set

$$x_h = \frac{1}{h}\int_0^h T(s)x \, \mathrm{d}s.$$

Strong continuity gives $x_h \to x$, while

$$\frac{T(t)x_h - x_h}{t} = \frac{1}{h}\left(\frac{1}{t}\int_h^{h+t} T(s)x \, \mathrm{d}s - \frac{1}{t}\int_0^t T(s)x \, \mathrm{d}s\right) \longrightarrow \frac{T(h)x - x}{h}.$$

Thus $x_h \in \mathrm{D}(A)$.

For $x \in \mathrm{D}(A)$,

$$\frac{T(h)T(t)x - T(t)x}{h} = T(t)\frac{T(h)x - x}{h} \longrightarrow T(t)Ax,$$

which proves invariance and commutation. The derivative at arbitrary t follows from the semigroup law; continuity of $s \mapsto T(s)Ax$ makes it a continuous derivative.

Finally, suppose $x_n \to x$ and $Ax_n \to y$. The fundamental theorem of calculus gives

$$T(t)x_n - x_n = \int_0^t T(s)Ax_n \, \mathrm{d}s.$$

Passing to the limit yields

$$T(t)x - x = \int_0^t T(s)y \, \mathrm{d}s.$$

Divide by t and let $t \downarrow 0$ to obtain $x \in \mathrm{D}(A)$ and $Ax = y$. $\square$

Definition 6.1.3. A C_0-semigroup is a *contraction semigroup* if $\|T(t)\| \leq 1$ for every $t \geq 0$.

6.2 Resolvents of generators

Lemma 6.2.1 (Laplace-transform resolvent)**.** Let $(T(t))$ be a contraction semigroup with generator A. For $\operatorname{Re} \lambda > 0$, define

$$R(\lambda, A)x = \int_0^\infty e^{-\lambda t} T(t)x \, \mathrm{d}t.$$

Then the Bochner integral is well defined,

$$R(\lambda, A) = (\lambda I - A)^{-1}, \qquad \|R(\lambda, A)\| \leq \frac{1}{\operatorname{Re} \lambda}.$$

For real $\lambda > 0$,

$$\|\lambda R(\lambda, A)\| \leq 1.$$

Proof. The norm estimate follows from

$$\int_0^\infty e^{-\operatorname{Re} \lambda t} \|T(t)x\| \, \mathrm{d}t \leq \frac{\|x\|}{\operatorname{Re} \lambda}.$$

For $s > 0$, the semigroup law and a change of variables give

$$\frac{T(s) - I}{s} R(\lambda, A)x = \frac{1}{s} \int_0^\infty e^{-\lambda t} \big(T(t+s) - T(t)\big)x \, \mathrm{d}t \longrightarrow \lambda R(\lambda, A)x - x.$$

Hence $R(\lambda, A)x \in \mathrm{D}(A)$ and $(\lambda I - A)R(\lambda, A)x = x$. If $x \in \mathrm{D}(A)$, integration by parts in $e^{-\lambda t} T(t)x$ gives $R(\lambda, A)(\lambda I - A)x = x$. Thus it is the inverse. $\square$

Corollary 6.2.2. The generator of a contraction semigroup is densely defined and closed, $(0, \infty) \subseteq \rho(A)$, and

$$\|\lambda(\lambda I - A)^{-1}\| \leq 1 \qquad (\lambda > 0).$$

6.3 The contraction Hille–Yosida theorem

Theorem 6.3.1 (Hille–Yosida theorem, contraction form)**.** Let $A\colon \mathrm{D}(A) \subseteq X \to X$ be linear. Then A generates a contraction C_0-semigroup if and only if

(a). A is densely defined and closed;

(b). $(0, \infty) \subseteq \rho(A)$;

(c). $\|\lambda(\lambda I - A)^{-1}\| \leq 1$ for every $\lambda > 0$.

Sufficiency. For $\lambda > 0$, define the bounded Yosida approximation

$$A_\lambda = \lambda A(\lambda I - A)^{-1} = \lambda^2(\lambda I - A)^{-1} - \lambda I.$$

Set

$$T_\lambda(t) = e^{tA_\lambda}.$$

Since $\|\lambda(\lambda I - A)^{-1}\| \leq 1$,

$$\|T_\lambda(t)\| = \left\| e^{-\lambda t} \exp\left(t\lambda^2(\lambda I - A)^{-1}\right) \right\| \leq 1.$$

For $x \in \mathrm{D}(A)$,

$$A_\lambda x = \lambda(\lambda I - A)^{-1} A x \longrightarrow Ax$$

as $\lambda \to \infty$. The resolvent identity shows that the bounded operators A_λ commute. Duhamel's formula therefore gives, for $x \in \mathrm{D}(A)$,

$$\|T_\lambda(t)x - T_\mu(t)x\| \leq \int_0^t \|(A_\lambda - A_\mu)x\| \, \mathrm{d}s = t\|(A_\lambda - A_\mu)x\|,$$

uniformly for t in compact intervals. Density of $\mathrm{D}(A)$ and the contraction bounds extend this Cauchy property to every $x \in X$. Define

$$T(t)x = \lim_{\lambda \to \infty} T_\lambda(t)x.$$

Uniform convergence on compact time intervals gives the semigroup law, strong continuity, and $\|T(t)\| \leq 1$. For $x \in \mathrm{D}(A)$,

$$T(t)x - x = \lim_{\lambda \to \infty} \int_0^t T_\lambda(s) A_\lambda x \, \mathrm{d}s = \int_0^t T(s)Ax \, \mathrm{d}s,$$

so the generator extends A. Equality of resolvents for one positive parameter, or closedness of A, shows that the generator is exactly A. Necessity was proved by the Laplace-transform lemma. □

Corollary 6.3.2 (Uniqueness and the abstract Cauchy problem)**.** A generator determines its C_0-semigroup uniquely. If $x_0 \in \mathrm{D}(A)$, then

$$u(t) = T(t)x_0$$

is the unique classical solution of

$$u'(t) = Au(t), \qquad u(0) = x_0.$$

Moreover, $\mathrm{D}(A^2)$ is a core for A and hence is dense in $\mathrm{D}(A)$ for the graph norm.

Proof. If u is another classical solution and T is generated by A, then for fixed t,

$$\frac{\mathrm{d}}{\mathrm{d}s}T(t-s)u(s) = -AT(t-s)u(s) + T(t-s)Au(s) = 0,$$

so $u(t) = T(t)u(0)$. For the core statement, apply $\lambda R(\lambda, A)$ to elements of $\mathrm{D}(A)$: these approximants lie in $\mathrm{D}(A^2)$ and converge in the graph norm as $\lambda \to \infty$. □

Theorem 6.3.3 (Euler approximation)**.** If A generates a contraction semigroup, then for every $x \in X$ and $t \geq 0$,

$$T(t)x = \lim_{n\to\infty} \left(I - \frac{t}{n}A\right)^{-n} x.$$

Proof. The expression is the exponential approximation associated with the Yosida operator at $\lambda = n/t$; the convergence follows from the construction in the Hille–Yosida proof. □

6.4 Exponentially bounded semigroups

Lemma 6.4.1. Every C_0-semigroup is exponentially bounded: there are $M \geq 1$ and $\omega \in \mathbb{R}$ such that

$$\|T(t)\| \leq Me^{\omega t} \qquad (t \geq 0).$$

Proof. Strong continuity and uniform boundedness give $M_0 = \sup_{0\leq t\leq 1} \|T(t)\| < \infty$. Writing $t = n + s$ with $n \in \mathbb{N}_0$ and $0 \leq s < 1$ gives $\|T(t)\| \leq M_0^{n+1} \leq Me^{\omega t}$ for suitable M, ω. □

Lemma 6.4.2 (Higher resolvent powers)**.** If $\|T(t)\| \leq Me^{\omega t}$ and A is its generator, then for $\operatorname{Re}\lambda > \omega$ and $n \geq 1$,

$$R(\lambda, A)^n x = \frac{1}{(n-1)!} \int_0^\infty t^{n-1} e^{-\lambda t} T(t)x \,\mathrm{d}t,$$

and therefore

$$\|R(\lambda, A)^n\| \leq \frac{M}{(\operatorname{Re}\lambda - \omega)^n}.$$

Proof. The formula for $n = 1$ is the Laplace representation. Induction, Fubini's theorem, and the semigroup law reduce the n-fold integral over $t_1 + \cdots + t_n = t$ to the simplex volume $t^{n-1}/(n-1)!$. The estimate is the Gamma-integral calculation. □

Theorem 6.4.3 (Hille–Yosida–Phillips theorem)**.** Let A be densely defined and closed. It generates a C_0-semigroup satisfying

$$\|T(t)\| \leq Me^{\omega t}$$

if and only if $(\omega, \infty) \subseteq \rho(A)$ and

$$\|R(\lambda, A)^n\| \leq \frac{M}{(\lambda - \omega)^n} \qquad (\lambda > \omega,\ n \geq 1).$$

Proof outline. Necessity is the preceding lemma. For sufficiency, shift the operator by ωI and apply the Yosida construction with the uniform power bounds. Those bounds replace the contraction estimate and yield uniformly bounded approximating semigroups after multiplication by $e^{-\omega t}$. Passing to the strong limit gives the required semigroup and growth estimate. □

6.5 Examples

Theorem 6.5.1 (Translation semigroup)**.** Let

$$X = C_0([0,\infty)) = \{f \in C([0,\infty)) : f(t) \to 0 \text{ as } t \to \infty\}$$

with the supremum norm, and define

$$(T(t)f)(x) = f(x+t).$$

Then $(T(t))$ is a contraction C_0-semigroup. Its generator is

$$Af = f',$$

with domain

$$\mathrm{D}(A) = \{f \in C_0([0,\infty)) : f \text{ is } C^1 \text{ and } f' \in C_0([0,\infty))\}.$$

Proof. Uniform continuity of every $f \in C_0([0,\infty))$ gives strong continuity. For $f \in \mathrm{D}(A)$, the difference quotient converges uniformly to f'. For the converse, if the uniform difference quotient converges to g, then

$$f(x+t) - f(x) = \int_0^t g(x+s)\, \mathrm{d}s,$$

so f is differentiable with $f' = g \in C_0$. □

Proposition 6.5.2 (Positive self-adjoint operators)**.** Let H be a Hilbert space and let A be densely defined, self-adjoint, and positive:

$$\langle Ax, x\rangle \geq 0 \qquad (x \in \mathrm{D}(A)).$$

Then $-A$ generates a contraction C_0-semigroup.

Proof. For $\lambda > 0$,

$$\|\lambda x + Ax\|\,\|x\| \geq \operatorname{Re}\langle(\lambda I + A)x, x\rangle \geq \lambda\|x\|^2,$$

so $\|\lambda(\lambda I + A)^{-1}\| \leq 1$ whenever the inverse exists. The spectral theorem, or the standard range argument for positive self-adjoint operators, gives $\operatorname{Ran}(\lambda I + A) = H$. Hille–Yosida applied to $-A$ gives the claim. The sign is essential: a positive operator itself generally produces growth rather than a contraction. □

6.6 Dissipative operators and Lumer–Phillips

Definition 6.6.1. For $x \in X$, the normalized duality set is

$$\mathcal{J}(x) = \{x^* \in X^* : x^*(x) = \|x\|^2,\ \|x^*\| = \|x\|\}.$$

It is nonempty by Hahn–Banach. A densely defined operator A is *accretive* if, for every $x \in \mathrm{D}(A)$, there is

$x^* \in \mathcal{J}(x)$ with

$$\operatorname{Re} x^*(Ax) \geq 0.$$

It is *dissipative* if $-A$ is accretive.

Proposition 6.6.2. An operator A is dissipative if and only if

$$\|(\lambda I - A)x\| \geq \lambda\|x\| \qquad (x \in \mathrm{D}(A),\ \lambda > 0).$$

Proof. If $x^* \in \mathcal{J}(x)$ and $\operatorname{Re} x^*(Ax) \leq 0$, then

$$\|(\lambda I - A)x\|\,\|x\| \geq \operatorname{Re} x^*((\lambda I - A)x) \geq \lambda\|x\|^2.$$

The converse follows by taking a supporting functional for x and letting $\lambda \downarrow 0$ in the norm inequality, or equivalently by the standard directional-derivative characterization of the norm. $\square$

Theorem 6.6.3 (Lumer–Phillips theorem). Let A be densely defined on a Banach space. Then A generates a contraction C_0-semigroup if and only if

(a). A is dissipative;

(b). $\operatorname{Ran}(\lambda_0 I - A) = X$ for some $\lambda_0 > 0$.

In this case the range condition holds for every $\lambda > 0$.

Proof. A contraction generator is dissipative by differentiating $\|T(t)x\| \leq \|x\|$ at $t = 0$ through a supporting functional, and its positive resolvent is onto. Conversely, dissipativity makes $\lambda I - A$ injective with inverse norm at most $1/\lambda$ on its range. The resolvent identity propagates surjectivity from one positive λ_0 to all positive λ by overlapping Neumann-series intervals. The operator is closed, and the contraction Hille–Yosida theorem applies. $\square$

Corollary 6.6.4. Let A be densely defined and closed on a Banach space. If both A and its Banach adjoint A^* are dissipative, then A generates a contraction C_0-semigroup.

Proof. Dissipativity makes $\operatorname{Ran}(I - A)$ closed. If it were not dense, Hahn–Banach would give $0 \neq f \in X^*$ annihilating it. Then $f \in \mathrm{D}(A^*)$ and $(I - A^*)f = 0$, contradicting the dissipative inequality for A^*. Hence the range is dense and closed, so it equals X. Apply Lumer–Phillips. $\square$

Definition 6.6.5. Let A be closed. A subspace $D \subseteq \mathrm{D}(A)$ is a *core* for A if the graph of $A|_D$ is dense in $\operatorname{Graph}(A)$, equivalently, if D is dense in $\mathrm{D}(A)$ for the graph norm $\|x\| + \|Ax\|$.

Theorem 6.6.6 (Invariant cores). Let A generate a C_0-semigroup $(T(t))$. Suppose $D \subseteq \mathrm{D}(A)$ is dense in X and $T(t)D \subseteq D$ for every $t \geq 0$. Then D is a core for A.

Proof. Let B be the graph closure of $A|_D$. Invariance and the orbit derivative show that $B \subseteq A$ and that B generates the restriction of the same semigroup to its closure. For sufficiently large λ, the Laplace resolvent maps the dense set D into the graph closure of $A|_D$. Since the resolvent is bounded and onto $\mathrm{D}(A)$ in the graph norm, that closure is all of $\mathrm{D}(A)$. $\square$

Theorem 6.6.7 (Gaussian heat semigroup). On $X = C_0(\mathbb{R}^d)$, define for $t > 0$

$$(T(t)f)(x) = \frac{1}{(4\pi t)^{d/2}} \int_{\mathbb{R}^d} e^{-|x-y|^2/(4t)} f(y)\, \mathrm{d}y, \qquad T(0) = I.$$

Then $(T(t))$ is a contraction C_0-semigroup. Its generator is the closure of

$$\Delta \colon C_c^\infty(\mathbb{R}^d) \to C_0(\mathbb{R}^d),$$

and $C_c^\infty(\mathbb{R}^d)$ is a core.

Proof. The Gaussian kernels form an approximate identity and satisfy $G_t * G_s = G_{t+s}$, which follows from Fourier transform or direct Gaussian integration. Positivity and unit mass give the contraction estimate, and approximate-identity convergence gives strong continuity on C_0. For $u \in C_c^\infty$, differentiation under the integral or Fourier transform gives $\frac{\mathrm{d}}{\mathrm{d}t} T(t)u = \Delta T(t)u = T(t)\Delta u$. The invariant-core theorem identifies the generator with the closure of Δ on C_c^∞. □

Remark 6.6.8. The same formula is not strongly continuous on all of $C_b(\mathbb{R}^d)$: functions in C_b need not be uniformly continuous. The correct standard spaces are $C_0(\mathbb{R}^d)$ or the bounded uniformly continuous functions.

Theorem 6.6.9 (Poisson semigroup). Let

$$P_t(x) = c_d \frac{t}{(t^2 + |x|^2)^{(d+1)/2}}, \qquad t > 0,$$

where c_d normalizes the integral to one. Then

$$T(t)f = P_t * f$$

defines a contraction C_0-semigroup on $C_0(\mathbb{R}^d)$. Its generator is

$$A = -(-\Delta)^{1/2},$$

so on a suitable core

$$A^2 = -\Delta.$$

Proof. The Fourier multiplier of P_t is $e^{-t|\xi|}$, so the semigroup property and contraction estimate follow. Differentiation at $t = 0$ gives the multiplier $-|\xi|$, which is $-(-\Delta)^{1/2}$. Squaring gives the multiplier $|\xi|^2$, corresponding to $-\Delta$. □

Chapter 7 Sobolev Spaces

7.1 One-dimensional Sobolev spaces

Definition 7.1.1. Let $I \subseteq \mathbb{R}$ be an open interval and let $1 \le p \le \infty$. A function $u \in L^p(I)$ belongs to $W^{1,p}(I)$ if there is $v \in L^p(I)$ such that

$$\int_I u\varphi' \, \mathrm{d}x = -\int_I v\varphi \, \mathrm{d}x \qquad (\varphi \in C_c^\infty(I)).$$

The function v is unique almost everywhere and is denoted by u'. The norm is

$$\|u\|_{W^{1,p}(I)} = \|u\|_{L^p(I)} + \|u'\|_{L^p(I)}.$$

We write $H^1(I) = W^{1,2}(I)$.

Theorem 7.1.2. The space $W^{1,p}(I)$ is Banach for every $1 \le p \le \infty$, separable for $1 \le p < \infty$, and reflexive for $1 < p < \infty$.

Proof. The map

$$u \longmapsto (u, u')$$

is an isometry from $W^{1,p}(I)$ into $L^p(I) \times L^p(I)$. Its range is closed: if $u_n \to u$ and $u_n' \to v$ in L^p, then for every test function

$$\int u\varphi' = \lim_n \int u_n\varphi' = -\lim_n \int u_n'\varphi = -\int v\varphi.$$

Completeness, separability, and reflexivity therefore follow from the corresponding product-space properties of L^p. □

Theorem 7.1.3 (One-dimensional fundamental theorem)**.** Let $u \in W^{1,p}(I)$. There is an absolutely continuous representative $\widetilde{u}$ such that

$$\widetilde{u}(x) - \widetilde{u}(y) = \int_y^x u'(t) \, \mathrm{d}t \qquad (x, y \in I).$$

In particular, every one-dimensional Sobolev function has a continuous representative.

Proof. Fix $y_0 \in I$ and define

$$v(x) = \int_{y_0}^x u'(t) \, \mathrm{d}t.$$

Then v is absolutely continuous and has weak derivative u'. Hence $u - v$ has weak derivative zero. The lemma below shows that $u - v$ is almost everywhere constant, so u agrees almost everywhere with $v +$

c. ☐

Lemma 7.1.4. If $f \in L^1_{\text{loc}}(I)$ and

$$\int_I f\varphi'\, dx = 0 \qquad (\varphi \in C_c^\infty(I)),$$

then f is almost everywhere constant on I.

Proof. Choose $\psi \in C_c^\infty(I)$ with $\int_I \psi = 1$. For any $\varphi \in C_c^\infty(I)$, the function $\varphi - (\int \varphi)\psi$ has integral zero and hence is the derivative of a compactly supported smooth function. Therefore

$$\int f\varphi = \left(\int f\psi\right)\left(\int \varphi\right),$$

which says that the distribution defined by f is the constant $c = \int f\psi$. ☐

Proposition 7.1.5 (Dual characterization of a weak derivative)**.** Let $u \in L^p(I)$ and $1 < p \leq \infty$. Then $u \in W^{1,p}(I)$ if and only if there is $C > 0$ such that

$$\left|\int_I u\varphi'\, dx\right| \leq C\|\varphi\|_{L^{p'}(I)} \qquad (\varphi \in C_c^\infty(I)),$$

with $p' = 1$ when $p = \infty$. The smallest such C is $\|u'\|_p$.

Proof. One direction is Hölder's inequality. Conversely, the map $\varphi \mapsto -\int u\varphi'$ is bounded on the dense subspace $C_c^\infty(I) \subset L^{p'}(I)$. Extend it to $L^{p'}$ and apply the L^p duality theorem to obtain $v \in L^p$ with $-\int u\varphi' = \int v\varphi$. ☐

Remark 7.1.6. At $p = 1$, the displayed estimate with the L^∞ norm controls the distributional derivative only as a finite measure and characterizes bounded variation. It does not by itself force that measure to have an L^1 density. This is why the theorem is stated for $p > 1$ (and for $p = \infty$ via $(L^1)^* = L^\infty$).

Remark 7.1.7. For $p = 1$, the absolutely continuous representative is exactly the classical one: a function belongs to $W^{1,1}(I)$ precisely when it is almost everywhere equal to an absolutely continuous function whose classical derivative lies in L^1. Merely bounded variation is weaker; BV functions may have jump or singular derivatives.

Theorem 7.1.8 (Lipschitz characterization)**.** A function $u \in L^\infty(I)$ belongs to $W^{1,\infty}(I)$ if and only if it has a Lipschitz representative. Moreover,

$$\text{Lip}(u) = \|u'\|_\infty.$$

Proof. The fundamental theorem gives $|u(x) - u(y)| \leq \|u'\|_\infty |x - y|$. Conversely, a Lipschitz function is absolutely continuous, differentiable almost everywhere, and its derivative is bounded by its Lipschitz constant. Integration by parts then identifies that derivative as the weak derivative. ☐

Theorem 7.1.9 (Difference-quotient criterion on $\mathbb{R}$)**.** Let $1 < p < \infty$ and $u \in L^p(\mathbb{R})$. Then $u \in W^{1,p}(\mathbb{R})$ if and only if there is $C > 0$ such that

$$\|u(\cdot + h) - u\|_p \leq C|h| \qquad (h \in \mathbb{R}).$$

The least admissible C equals $\|u'\|_p$.

Proof. If $u \in W^{1,p}$, then for the absolutely continuous representative,

$$u(x+h) - u(x) = \int_0^h u'(x+s)\,\mathrm{d}s.$$

Minkowski's integral inequality gives the estimate with $C = \|u'\|_p$. Conversely, for $\varphi \in C_c^\infty(\mathbb{R})$,

$$\int u(x)\frac{\varphi(x-h) - \varphi(x)}{h}\,\mathrm{d}x = \int \frac{u(x+h) - u(x)}{h}\varphi(x)\,\mathrm{d}x.$$

The right-hand side is bounded by $C\|\varphi\|_{p'}$. Letting $h \to 0$ and using the preceding dual characterization yields a weak derivative in L^p. For $p = 1$, the analogous uniform translation bound characterizes bounded variation rather than $W^{1,1}$, so the endpoint cannot be inserted without an additional absolute-continuity condition. □

7.2 Extension and smooth approximation

Theorem 7.2.1 (Extension from an interval)**.** Let I be an interval and $1 \leq p \leq \infty$. There is a bounded linear operator

$$E\colon W^{1,p}(I) \to W^{1,p}(\mathbb{R})$$

such that $Eu = u$ almost everywhere on I and

$$\|Eu\|_{W^{1,p}(\mathbb{R})} \leq C_I\|u\|_{W^{1,p}(I)}.$$

For a bounded interval, Eu may be chosen with support in a fixed bounded neighbourhood of I.

Construction. For a half-line, reflect across the endpoint on a short adjacent interval and then multiply by a fixed smooth cutoff. Reflection preserves the L^p norms of both the function and its weak derivative up to a fixed factor. For a bounded interval, reflect across both endpoints, use a partition of unity near the two ends, and multiply by a cutoff supported in a slightly larger interval. The product rule gives the stated bound. □

Lemma 7.2.2 (Mollification of Sobolev functions)**.** Let $u \in W^{1,p}(\mathbb{R})$ and let $\rho \in C_c^\infty(\mathbb{R})$. Then

$$\rho * u \in W^{1,p}(\mathbb{R}), \qquad (\rho * u)' = \rho * u'.$$

Proof. For a test function φ, Fubini's theorem and the definition of the weak derivative give

$$\int (\rho * u)\varphi' = \int u\,(\widetilde{\rho} * \varphi') = -\int u'\,(\widetilde{\rho} * \varphi) = -\int (\rho * u')\varphi,$$

where $\widetilde{\rho}(x) = \rho(-x)$. Young's inequality gives the L^p bounds. □

Theorem 7.2.3 (Smooth approximation)**.** For $1 \leq p < \infty$, $C_c^\infty(\mathbb{R})$ is dense in $W^{1,p}(\mathbb{R})$. More generally, smooth functions are dense in $W^{1,p}(I)$, and on a bounded interval they may be obtained by extending, mollifying, and restricting.

Proof. Let ρ_n be mollifiers. Then

$$\rho_n * u \to u, \qquad \rho_n * u' \to u'$$

in L^p, so $\rho_n * u \to u$ in $W^{1,p}$. Multiplying by cutoffs that tend to one produces compact support while the product-rule error tends to zero. For an interval, first apply the bounded extension operator. □

7.3 One-dimensional Sobolev embeddings

Theorem 7.3.1 (Sobolev embedding on an interval). Let I be an interval of finite length and $1 \le p \le \infty$. Every $u \in W^{1,p}(I)$ has a continuous representative and

$$\|u\|_{L^\infty(I)} \le C_I \|u\|_{W^{1,p}(I)}.$$

Thus $W^{1,p}(I) \hookrightarrow L^\infty(I)$ continuously.

If I is bounded, then

(a). $W^{1,p}(I) \hookrightarrow C(\bar{I})$ compactly for $1 < p \le \infty$;

(b). $W^{1,1}(I) \hookrightarrow L^2(I)$ compactly.

Proof. For $1 < p < \infty$, apply the fundamental theorem to $G(u) = |u|^{p-1}u$ (or a smooth approximation in the complex case):

$$|u(x)|^p \le p \int_I |u|^{p-1}|u'| + \frac{1}{|I|}\|u\|_p^p,$$

and Hölder's inequality gives the stated L^∞ estimate. The endpoint $p = 1$ follows directly from $|u(x)| \le |I|^{-1}\|u\|_1 + \|u'\|_1$, and $p = \infty$ is immediate.

For $p > 1$, a bounded set in $W^{1,p}(I)$ is uniformly bounded and satisfies

$$|u(x) - u(y)| \le \|u'\|_p |x - y|^{1/p'},$$

so Arzelà–Ascoli gives compactness in $C(\bar{I})$. For $p = 1$, use the extension operator. A bounded family is bounded in both L^1 and L^∞, and translations satisfy $\|u(\cdot + h) - u\|_1 \le C|h|$. The Fréchet–Kolmogorov compactness criterion gives relative compactness in L^1; interpolation with the uniform L^∞ bound yields relative compactness in L^2. □

Corollary 7.3.2 (Decay on an unbounded interval). Let I be unbounded and let $1 \le p < \infty$. If $u \in W^{1,p}(I)$, then its continuous representative satisfies

$$u(x) \longrightarrow 0$$

at every infinite endpoint of I.

Proof. Approximate u in $W^{1,p}(I)$ by compactly supported smooth functions. The Sobolev L^∞ estimate makes this approximation uniform, so u is a uniform limit of functions vanishing at infinity. □

Remark 7.3.3. On an unbounded interval, the embedding $W^{1,p}(I) \hookrightarrow L^\infty(I)$ is not compact. Translates of one fixed nonzero compactly supported smooth function are bounded in $W^{1,p}$ but have no uniformly convergent subsequence.

7.4 Product and chain rules

Proposition 7.4.1 (One-dimensional differentiation rules). Let $1 \le p \le \infty$.

(a). If $u, v \in W^{1,p}(I) \cap L^\infty(I)$, then $uv \in W^{1,p}(I)$ and

$$(uv)' = u'v + uv'.$$

The integration-by-parts identity

$$\int_a^b u'v = u(b)v(b) - u(a)v(a) - \int_a^b uv'$$

holds for bounded $I = (a, b)$.

(b). If $G \in C^1(\mathbb{R})$, $G(0) = 0$, and G' is bounded, then $G \circ u \in W^{1,p}(I)$ and

$$(G \circ u)' = (G' \circ u)u'.$$

Proof. Use smooth approximation in $W^{1,p}$ together with the one-dimensional L^∞ embedding on bounded subintervals. The classical identities hold for the approximants, and the products converge in L^p. For the chain rule, first handle bounded G' directly and pass to the limit; local truncation handles the corresponding local version. □

7.5 Higher-order spaces, traces, and negative norms

Definition 7.5.1. For an integer $m \ge 1$,

$$W^{m,p}(I) = \{u \in L^p(I) : u^{(j)} \in L^p(I),\ 1 \le j \le m\},$$

with norm

$$\|u\|_{W^{m,p}(I)} = \sum_{j=0}^{m} \|u^{(j)}\|_p.$$

Equivalent ℓ^p combinations of these terms give the same topology. When $p = 2$ we write $H^m(I) = W^{m,2}(I)$.

Definition 7.5.2. For $1 \le p < \infty$,

$$W_0^{1,p}(I) = \overline{C_c^\infty(I)}^{W^{1,p}(I)}.$$

Theorem 7.5.3 (Trace characterization). Let $I = (a, b)$ be bounded. Then

$$W_0^{1,p}(I) = \{u \in W^{1,p}(I) : u(a) = u(b) = 0\},$$

where the boundary values are taken from the continuous representative.

Proof. Trace continuity follows from the Sobolev embedding, so every limit of compactly supported smooth functions has zero endpoint values. Conversely, if u has zero endpoints, multiply it by cutoffs that equal one away from successively smaller endpoint intervals. The zero trace and the fundamental theorem make the cutoff errors tend to zero in $W^{1,p}$; mollification then gives C_c^∞ approximation. □

Theorem 7.5.4 (Poincaré inequality)**.** For a bounded interval I and $1 \le p < \infty$, there is $C_I > 0$ such that

$$\|u\|_p \le C_I \|u'\|_p \qquad (u \in W_0^{1,p}(I)).$$

The same estimate holds for zero-trace functions in $W^{1,\infty}(I)$. Consequently, $\|u'\|_p$ is an equivalent norm on $W_0^{1,p}(I)$.

Proof. For the continuous representative and $u(a) = 0$,

$$|u(x)| \le \int_a^x |u'(t)| \, dt.$$

Hölder's inequality and integration in x give the result. □

Definition 7.5.5. For $m \ge 1$, $W_0^{m,p}(I)$ is the closure of $C_c^\infty(I)$ in $W^{m,p}(I)$. For $1 < p < \infty$, the dual of $W_0^{1,p}(I)$ is denoted

$$W^{-1,p'}(I),$$

and the dual of $H_0^1(I)$ is denoted $H^{-1}(I)$.

Theorem 7.5.6 (Representation of $W^{-1,p'}$)**.** Let $1 < p < \infty$. A functional F belongs to $W^{-1,p'}(I)$ if and only if there are $f_0, f_1 \in L^{p'}(I)$ such that

$$\langle F, u \rangle = \int_I f_0 u \, dx + \int_I f_1 u' \, dx \qquad (u \in W_0^{1,p}(I)).$$

Moreover,

$$\|F\|_{W^{-1,p'}} = \inf\{\|f_0\|_{p'} + \|f_1\|_{p'} : (f_0, f_1) \text{ represents } F\}$$

up to the harmless choice of an equivalent product norm. If I is bounded, Poincaré's inequality allows a representation with $f_0 = 0$.

Proof. Embed $W_0^{1,p}(I)$ isometrically as the closed subspace

$$G = \{(u, u') : u \in W_0^{1,p}(I)\} \subset L^p(I) \times L^p(I).$$

The functional F induces a bounded functional on G. Hahn–Banach extends it to the product, and L^p duality represents that extension by $(f_0, f_1) \in L^{p'} \times L^{p'}$. Conversely, Hölder's inequality makes every such formula continuous. On a bounded interval, use the derivative embedding $u \mapsto u'$ and Poincaré's inequality instead. □

7.6 Sobolev spaces in several variables

Definition 7.6.1. Let $\Omega \subseteq \mathbb{R}^N$ be open. A function $u \in L^p(\Omega)$ belongs to $W^{1,p}(\Omega)$ if, for each $1 \le i \le N$, there is $g_i \in L^p(\Omega)$ satisfying

$$\int_\Omega u\,\partial_i\varphi\,\mathrm{d}x = -\int_\Omega g_i\varphi\,\mathrm{d}x \qquad (\varphi \in C_c^\infty(\Omega)).$$

We write $\partial_i u = g_i$ and $\nabla u = (\partial_1 u, \dots, \partial_N u)$. The norm

$$\|u\|_{W^{1,p}(\Omega)} = \|u\|_p + \sum_{i=1}^N \|\partial_i u\|_p$$

may be replaced by any equivalent finite-dimensional combination of these terms.

Proposition 7.6.2. The space $W^{1,p}(\Omega)$ is Banach for every $1 \le p \le \infty$, separable for $1 \le p < \infty$, and reflexive for $1 < p < \infty$. The space $H^1(\Omega)$ is Hilbert under

$$\langle u, v\rangle_{H^1} = \int_\Omega u\bar{v}\,\mathrm{d}x + \sum_{i=1}^N \int_\Omega \partial_i u\,\overline{\partial_i v}\,\mathrm{d}x.$$

Proof. Use the closed graph of $u \mapsto (u, \partial_1 u, \dots, \partial_N u)$ in a finite product of L^p spaces, exactly as in one dimension. □

Proposition 7.6.3 (Closedness of weak differentiation)**.** Suppose $u_n \to u$ in $L^p(\Omega)$ and $\partial_i u_n \to g_i$ in $L^p(\Omega)$ for every i. Then $u \in W^{1,p}(\Omega)$ and $\partial_i u = g_i$. The same conclusion holds when all convergences are weak in the corresponding L^p spaces.

Definition 7.6.4. For open sets $\omega, \Omega \subseteq \mathbb{R}^N$, the notation $\omega \Subset \Omega$ means that $\overline{\omega}$ is compact and contained in Ω.

Lemma 7.6.5 (Local mollification)**.** If $u \in W^{1,p}(\Omega)$ and $\omega \Subset \Omega$, then for sufficiently small ε,

$$u_\varepsilon = \rho_\varepsilon * u$$

is smooth on ω and

$$\partial_i u_\varepsilon = \rho_\varepsilon * \partial_i u \quad \text{on } \omega.$$

Theorem 7.6.6 (Meyers–Serrin density theorem)**.** Let $1 \le p < \infty$. Then

$$C^\infty(\Omega) \cap W^{1,p}(\Omega)$$

is dense in $W^{1,p}(\Omega)$. If $\Omega = \mathbb{R}^N$, the approximants may be chosen in $C_c^\infty(\mathbb{R}^N)$. On a general open set, $C_c^\infty(\Omega)$ is dense only in the smaller space $W_0^{1,p}(\Omega)$, not in all of $W^{1,p}(\Omega)$.

Proof sketch. Choose an exhaustion $\Omega_1 \Subset \Omega_2 \Subset \cdots$ and a smooth partition of unity subordinate to annular pieces. Mollify each localized piece at a scale smaller than its distance from the boundary and choose the scales so that the sum of the $W^{1,p}$ errors is finite and tends to zero. On $\mathbb{R}^N$, first cut off at large radius and then mollify. □

Theorem 7.6.7 (Distribution and translation criteria)**.** Let $1 < p < \infty$ and $u \in L^p(\mathbb{R}^N)$. The following are equivalent:

(a). $u \in W^{1,p}(\mathbb{R}^N)$;

(b). for each i there is C_i such that

$$\left| \int_{\mathbb{R}^N} u\, \partial_i \varphi \, \mathrm{d}x \right| \le C_i \|\varphi\|_{p'} \qquad (\varphi \in C_c^\infty);$$

(c). there is $C > 0$ such that

$$\|u(\cdot + h) - u\|_p \le C|h| \qquad (h \in \mathbb{R}^N).$$

For $p = 1$, the analogous bounded distributional-derivative or translation condition characterizes BV, not necessarily $W^{1,1}$.

Proposition 7.6.8 (Differentiation rules in $\mathbb{R}^N$)**.** Let $1 \le p \le \infty$.

(a). If $u, v \in W^{1,p}(\Omega) \cap L^\infty(\Omega)$, then $uv \in W^{1,p}(\Omega)$ and

$$\partial_i(uv) = u\, \partial_i v + v\, \partial_i u.$$

(b). If $G \in C^1(\mathbb{R})$, $G(0) = 0$, and G' is bounded, then $G \circ u \in W^{1,p}(\Omega)$ and

$$\partial_i(G \circ u) = (G' \circ u)\partial_i u.$$

(c). Let $\Phi \colon \Omega' \to \Omega$ be a C^1 bi-Lipschitz diffeomorphism whose derivative and inverse derivative are bounded. If $u \in W^{1,p}(\Omega)$, then $u \circ \Phi \in W^{1,p}(\Omega')$ and

$$\nabla(u \circ \Phi) = D\Phi^{\mathsf{T}}(\nabla u) \circ \Phi$$

almost everywhere.

(d). If $\eta \in C_c^\infty(\Omega)$ and $u \in W^{1,p}_{\mathrm{loc}}(\Omega)$, then $\eta u \in W^{1,p}(\Omega)$ and the product rule holds.

Definition 7.6.9. For an integer $m \ge 1$, a multi-index α, and $1 \le p \le \infty$,

$$W^{m,p}(\Omega) = \{u \in L^p(\Omega) : D^\alpha u \in L^p(\Omega) \text{ for all } |\alpha| \le m\}.$$

It is equipped with

$$\|u\|_{W^{m,p}} = \sum_{|\alpha| \le m} \|D^\alpha u\|_p.$$

Equivalent finite-dimensional combinations of these terms define the same norm topology; for $p = 2$ this is the Hilbert space $H^m(\Omega) = W^{m,2}(\Omega)$.

Chapter 8 Calculus of Variations and Distributions

8.1 Quadratic minimization

Theorem 8.1.1 (Coercive quadratic minimization)**.** Let V be a real Banach space and let $a\colon V \times V \to \mathbb{R}$ be continuous, symmetric, and coercive: there are $M, \alpha > 0$ such that

$$|a(u,v)| \le M\|u\|\,\|v\|, \qquad a(v,v) \ge \alpha\|v\|^2.$$

For $\ell \in V^*$ define

$$J_\ell(v) = \frac{1}{2}a(v,v) - \ell(v).$$

If $K \subseteq V$ is nonempty, closed, and convex, then J_ℓ has a unique minimizer $u \in K$. The solution map $\ell \mapsto u$ is Lipschitz continuous. It is linear when K is a linear subspace.

Proof. The form a is an inner product, and $\|v\|_a = \sqrt{a(v,v)}$ is equivalent to the original norm. Since V is complete in the original norm, it is complete in $\|\cdot\|_a$ and is therefore a Hilbert space. Riesz representation gives a unique $c \in V$ such that

$$\ell(v) = a(c,v).$$

Then

$$J_\ell(v) = \frac{1}{2}\|v-c\|_a^2 - \frac{1}{2}\|c\|_a^2.$$

Thus the minimizer is the metric projection $P_K^a c$ in the Hilbert norm $\|\cdot\|_a$. Existence, uniqueness, and nonexpansiveness follow from the projection theorem. Since

$$\|c_1 - c_2\|_a \le \alpha^{-1/2}\|\ell_1 - \ell_2\|_{V^*}$$

up to the norm-equivalence constants, the solution map is Lipschitz. If K is a subspace, the orthogonal projection is linear. □

Corollary 8.1.2 (Variational inequality)**.** The minimizer $u \in K$ is characterized by

$$a(u, v-u) \ge \ell(v-u) \qquad (v \in K).$$

If K is a linear subspace, this is equivalent to the variational equation

$$a(u,v) = \ell(v) \qquad (v \in K).$$

Proof. For $v \in K$ and $0 < t \le 1$, minimality of u applied to $u + t(v-u)$ gives

$$0 \le \frac{J(u+t(v-u)) - J(u)}{t} = a(u, v-u) - \ell(v-u) + \frac{t}{2} a(v-u, v-u).$$

Let $t \downarrow 0$. Conversely, expanding $J(v) - J(u)$ and using the inequality gives nonnegativity. If K is a subspace, test with $u \pm v$ to obtain equality. $\square$

Remark 8.1.3. The first variation of $J(v) = \frac{1}{2}a(v,v) - \ell(v)$ in the direction w is

$$J'(v)w = a(v,w) - \ell(w).$$

The additional term $\frac{1}{2}a(w,w)$ in the finite difference $J(v+w) - J(v)$ is second order. Quadratic minimization on a subspace therefore produces a linear variational equation, while nonlinear functionals generally produce nonlinear Euler–Lagrange equations.

8.2 Lax–Milgram and Stampacchia

Theorem 8.2.1 (Lax–Milgram lemma)**.** Let H be a real or complex Hilbert space and let $a \colon H \times H \to \mathbb{K}$ be a continuous bilinear or sesquilinear form. Assume coercivity:

$$\operatorname{Re} a(v,v) \ge \alpha \|v\|^2 \qquad (v \in H)$$

for some $\alpha > 0$. Then, for every $\ell \in H^*$, there is a unique $u \in H$ such that

$$a(u,v) = \ell(v) \qquad (v \in H).$$

Moreover,

$$\|u\| \le \frac{1}{\alpha}\|\ell\|,$$

and the map $\ell \mapsto u$ is bounded and linear.

Proof. Riesz representation defines a bounded operator $A \in \mathcal{B}(H,H)$ by

$$a(u,v) = \langle Au, v \rangle .$$

Coercivity gives

$$\alpha \|u\|^2 \le \operatorname{Re} \langle Au, u \rangle \le \|Au\| \, \|u\|,$$

so A is injective, bounded below, and has closed range. If $y \perp \operatorname{Ran}(A)$, then $A^* y = 0$ and

$$\alpha \|y\|^2 \le \operatorname{Re} a(y,y) = \operatorname{Re} \langle Ay, y \rangle = \operatorname{Re} \langle y, A^* y \rangle = 0,$$

so $y = 0$. Thus the range is dense and closed, hence all of H. Solve $Au = f$, where f is the Riesz representative of ℓ. The estimate follows from the lower bound. $\square$

Theorem 8.2.2 (Stampacchia variational inequality)**.** Let H be a Hilbert space, let $K \subseteq H$ be nonempty, closed, and convex, and let $a\colon H \times H \to \mathbb{K}$ be continuous and coercive. For every $\ell \in H^*$ there is a unique $u \in K$ such that

$$\operatorname{Re} a(u, v-u) \geq \operatorname{Re} \ell(v-u) \qquad (v \in K).$$

The solution depends Lipschitz continuously on ℓ.

Proof. Let A and f be the Riesz representatives of a and ℓ. For $\rho > 0$, define

$$\Phi(u) = P_K(u - \rho(Au - f)).$$

Using nonexpansiveness of P_K, continuity of A, and coercivity, one finds for sufficiently small $\rho > 0$ that

$$\|\Phi(u) - \Phi(v)\|^2 \leq \big(1 - 2\rho\alpha + \rho^2\|A\|^2\big)\|u - v\|^2.$$

Thus Φ is a contraction and has a unique fixed point $u \in K$. The projection characterization of this fixed point is exactly the variational inequality. Comparing the inequalities for two right-hand sides and using coercivity gives

$$\alpha\|u_1 - u_2\|^2 \leq \|\ell_1 - \ell_2\|\,\|u_1 - u_2\|,$$

which proves Lipschitz dependence. □

8.3 Test functions and weak derivatives

Definition 8.3.1. Let $\Omega \subseteq \mathbb{R}^N$ be open. The test-function space is

$$\mathcal{D}(\Omega) = C_c^\infty(\Omega).$$

The space $L^1_{\mathrm{loc}}(\Omega)$ consists of measurable functions that are integrable over every compact subset of Ω.

Definition 8.3.2. Let $u \in L^1_{\mathrm{loc}}(\Omega)$ and let α be a multi-index. A function $v \in L^1_{\mathrm{loc}}(\Omega)$ is the weak derivative $D^\alpha u$ if

$$\int_\Omega v\varphi \,\mathrm{d}x = (-1)^{|\alpha|}\int_\Omega uD^\alpha\varphi \,\mathrm{d}x \qquad (\varphi \in \mathcal{D}(\Omega)).$$

8.4 Distributions

Definition 8.4.1. A *distribution* on Ω is a linear functional $T\colon \mathcal{D}(\Omega) \to \mathbb{K}$ such that, for every compact $K \subseteq \Omega$, there are $C_K > 0$ and $m_K \in \mathbb{N}_0$ with

$$|T(\varphi)| \leq C_K \max_{|\alpha| \leq m_K} \sup_{x \in K} |D^\alpha\varphi(x)|$$

for every $\varphi \in \mathcal{D}(\Omega)$ supported in K. The space of all distributions is denoted by $\mathcal{D}'(\Omega)$.

Remark 8.4.2 (Topology). A sequence φ_n converges to φ in $\mathcal{D}(\Omega)$ if there is one compact $K \Subset \Omega$ containing all supports and

$$\sup_{x\in K} |D^\alpha(\varphi_n - \varphi)(x)| \longrightarrow 0$$

for every multi-index α. The full test-function topology is an inductive-limit topology and is generally not metrizable. A sequence $T_n \in \mathcal{D}'(\Omega)$ converges distributionally to T if

$$T_n(\varphi) \longrightarrow T(\varphi) \qquad (\varphi \in \mathcal{D}(\Omega)).$$

Proposition 8.4.3 (Regular distributions). Every $u \in L^1_{\mathrm{loc}}(\Omega)$ defines a distribution T_u by

$$T_u(\varphi) = \int_\Omega u\varphi \, \mathrm{d}x.$$

The map $u \mapsto T_u$ is injective modulo equality almost everywhere.

Proof. If $\operatorname{supp} \varphi \subseteq K$, then

$$|T_u(\varphi)| \leq \|u\|_{L^1(K)} \|\varphi\|_{L^\infty(K)}.$$

If $T_u = 0$, testing against mollified cutoffs and using the fundamental lemma of the calculus of variations gives $u = 0$ almost everywhere. □

Example 8.4.4 (Dirac delta). For $a \in \Omega$, the Dirac distribution is

$$\delta_a(\varphi) = \varphi(a).$$

It is not a regular distribution.

Proof. Suppose $\delta_a = T_u$ for some $u \in L^1_{\mathrm{loc}}$. Choose $\psi \in C_c^\infty(B(0,1))$ with $0 \leq \psi \leq 1$ and $\psi(0) = 1$, and set $\psi_k(x) = \psi(k(x-a))$. Then $\delta_a(\psi_k) = 1$, whereas

$$\left| \int u\psi_k \right| \leq \int_{B(a,1/k)} |u(x)| \, \mathrm{d}x \longrightarrow 0$$

by absolute continuity of the Lebesgue integral, a contradiction. □

Definition 8.4.5 (Distributional derivatives). For $T \in \mathcal{D}'(\Omega)$ and a multi-index α, define

$$D^\alpha T(\varphi) = (-1)^{|\alpha|} T(D^\alpha \varphi).$$

Then $D^\alpha T$ is again a distribution.

Proposition 8.4.6. If $u \in C^{|\alpha|}(\Omega)$, then

$$D^\alpha T_u = T_{D^\alpha u}.$$

More generally, when $u \in L^1_{\mathrm{loc}}$, the regular distribution T_v is $D^\alpha T_u$ exactly when v is the weak derivative $D^\alpha u$.

Proof. Repeated integration by parts against compactly supported test functions gives

$$D^\alpha T_u(\varphi) = (-1)^{|\alpha|} \int u D^\alpha \varphi = \int D^\alpha u \, \varphi.$$

The final statement is precisely the definition of weak differentiation. □